

Combinatorics

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Graph Theory

- Graphs & digraphs
- Paths & connectivity
- Trees & spanning trees
- Bipartite graphs
- Matchings & Hall's theorem
- Planarity & coloring
- Network flows

Languages & Computation

- Alphabets & formal languages
- Regular expressions
- Finite automata (DFA, NFA)
- Pumping lemma
- Context-free grammars
- Pushdown automata
- Turing machines
- Decidability & complexity

Combinatorics & Recurrences

- Counting principles
- Permutations & combinations
- Inclusion-exclusion
- Partitions & Stirling numbers
- Generating functions
- Recurrence relations
- Asymptotic analysis

Combinatorics

*“It is better to solve one problem five different ways,
than to solve five problems one way.”*

— George Pólya



George Pólya



Abraham de
Moivre



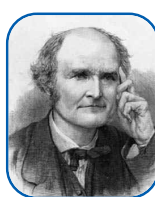
Leonhard Euler



Gottfried
Wilhelm
Leibniz



Herbert Wilf



Arthur Cayley



Augustin-Louis
Cauchy

What is Combinatorics?

Definition 1: *Combinatorics* is the branch of mathematics that studies the existence, enumeration, and structure of *finite* discrete objects.

Three fundamental questions arise in every combinatorial problem:

Existence

Does an arrangement of the required kind *exist* at all?

Example: Does a graph with 5 vertices where every vertex has exactly 3 neighbors exist?

Enumeration

How many such arrangements are there?

Example: How many 4-digit PIN codes are possible?

Optimization

Which arrangement is *best* by some criterion?

Example: What is the shortest tour visiting all n cities exactly once?

Basic Counting Principles

“Music is the pleasure the human mind experiences from counting without being aware that it is counting.”

— Gottfried Wilhelm Leibniz

Basic Counting Rules

Addition (*either/or*)

If S splits into *disjoint* parts $S_1, \dots, S_k$, count each part separately: $|S| = |S_1| + \dots + |S_k|$

Example: 26 vowels + 20 consonants = 46 letters.

Subtraction (complement)

Count what you *don't* want and subtract from the universe U : $|\overline{S}| = |U| - |S|$

Example: 8-bit strings *with* a leading zero: $2^8 - 2^7 = 128$.

Multiplication (*and*)

For *independent* sequential choices of sizes $n_1, \dots, n_k$, multiply: $|S| = n_1 \cdot n_2 \cdot \dots \cdot n_k$

Example: $26 \times 10 = 260$ two-character codes (letter then digit).

Bijection (one-to-one)

If S and T are in *bijective correspondence*, they have the same size: $|S| = |T|$

Example: $\binom{n}{k} = \binom{n}{n-k}$ via $A \mapsto [n] \setminus A$.

Addition Principle

Definition 2: We say a finite set S is *partitioned* into *parts* $S_1, \dots, S_k$ if the parts are pairwise disjoint and their union is S . In other words, $S_i \cap S_j = \emptyset$ for $i \neq j$ and $S_1 \cup S_2 \cup \dots \cup S_k = S$. In that case:

$$|S| = |S_1| + |S_2| + \dots + |S_k|$$

Example: Let S be the set of students attending the combinatorics lecture. It can be partitioned into parts S_1 and S_2 where

S_1 = set of students that *like* easy examples.

S_2 = set of students that *don't like* easy examples.

If $|S_1| = 22$ and $|S_2| = 8$, then we can conclude $|S| = |S_1| + |S_2| = 30$.

Multiplication Principle

Definition 3: If S is a finite set that is the *product* of $S_1, \dots, S_k$, that is, $S = S_1 \times \dots \times S_k$, then

$$|S| = |S_1| \times \dots \times |S_k|$$

Example: A *license plate* consists of 3 letters followed by 3 digits. The letters are chosen from $\{A, \dots, Z\}$ (26 choices each) and the digits from $\{0, \dots, 9\}$ (10 choices each). Since each position is filled independently:

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 = 26^3 \cdot 10^3 = 17,576,000$$

Subtraction Principle

Definition 4: Let S be a subset of a finite set T . We define the *complement* of S as $\overline{S} = T \setminus S$. Then

$$|\overline{S}| = |T| - |S|$$

Example: Let T be the set of all students at a university ($|T| = 23905$). Let $S \subseteq T$ be the set of students studying *neither* mathematics nor computer science ($|S| = 20178$). The complement $\overline{S} = T \setminus S$ consists of students studying mathematics or computer science:

$$|\overline{S}| = |T| - |S| = 23905 - 20178 = 3727$$

Bijection Principle

Definition 5: If S and T are sets, then

$$|S| = |T| \iff \text{there exists a bijection between } S \text{ and } T$$

Example: The number of k -subsets of $[n]$ equals the number of $(n - k)$ -subsets: $\binom{n}{k} = \binom{n}{n-k}$. The map $S \mapsto [n] \setminus S$ is a bijection from $\binom{[n]}{k}$ to $\binom{[n]}{n-k}$: each k -subset maps to its $(n - k)$ -element complement, and the map is its own inverse.

Note: The bijection principle works for *infinite* sets too — this is the foundation of Cantor's theory of cardinality.

Pigeonhole Principle

Definition 6: Let $S_1, \dots, S_k$ be finite sets that are pairwise disjoint and $|S_1| + |S_2| + \dots + |S_k| = n$.

$$\exists i \in \{1, \dots, k\} : |S_i| \geq \left\lfloor \frac{n}{k} \right\rfloor \quad \text{and} \quad \exists j \in \{1, \dots, k\} : |S_j| \leq \left\lceil \frac{n}{k} \right\rceil$$

Example: Assume there are 5 holes in the wall where pigeons nest. Say there is a set S_i of pigeons nesting in hole i . Assume there are $n = 17$ pigeons in total. Then we know:

- There is some hole with at least $d = 4$ pigeons.
- There is some hole with at most $b = 3$ pigeons.

Birthday attack: a hash function maps arbitrary inputs to n -bit digests. With $2^n + 1$ inputs, at least two must collide — guaranteed by the pigeonhole principle. More subtly, with only $\approx 2^{n/2}$ random inputs, collision probability already exceeds 50%. This is the *birthday bound* and sets the security level of hash functions in cryptography.

Double Counting

If we count the same quantity in *two different ways*, then this gives us a (perhaps non-trivial) identity.

Example (Handshaking Lemma): Assume there are n people at a party and everybody will shake hands with everybody else. How many handshakes will occur? We count this number in two ways:

1. Every person shakes $n - 1$ hands and there are n people. However, two people are involved in a handshake so if we just multiply $n \cdot (n - 1)$, then every handshake is counted twice. The total number of handshakes is therefore $\frac{n \cdot (n-1)}{2}$.
2. We number the people from 1 to n . To avoid counting a handshake twice, we count for person i only the handshakes with persons of lower numbers. Then the total number of handshakes is:

$$\sum_{i=1}^n (i-1) = \sum_{i=0}^{\{n-1\}} i = \sum_{i=1}^{n-1} i$$

The identity we obtain is therefore: $\sum_{i=1}^{n-1} i = \frac{n \cdot (n-1)}{2}$

Arrangements, Permutations, Combinations

*“As for everything else, so for a mathematical theory:
beauty can be perceived but not explained.”*

— Arthur Cayley

Ordered Arrangements

Definition 7: Denote by $[n] = \{1, \dots, n\}$ the set of natural numbers from 1 to n .

Hereinafter, let X be a finite set.

Definition 8: An *ordered arrangement* of n elements of X is a *map* $s : [n] \rightarrow X$.

- Here, $[n]$ is the *domain* of s , and $s(i)$ is the *image* of $i \in [n]$ under s .
- The set $\{x \in X \mid s(i) = x \text{ for some } i \in [n]\}$ is the *range* of s .

Other common names for ordered arrangements are:

- *string* (or *word*), e.g. “Banana”
- *sequence*, e.g. “0815422372”
- *tuple*, e.g. $(3, 5, 2, 5, 8)$

Example:

i	1	2	3	4	5	6	7
$s(i)$							

Permutations

Definition 9: A *permutation* of X is a *bijective* map $\pi : [n] \rightarrow X$.

Usually, $X = [n]$, and the set of all permutations of $[n]$ is denoted by S_n .

Example:

i	1	2	3	4	5	6	7
$\pi(i)$	2	7	1	3	5	4	6

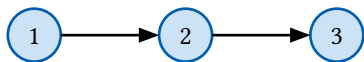
Definition 10: *k-permutation* of X is an ordered arrangement of k *distinct* elements of X , that is, an *injective* map $\pi : [k] \rightarrow X$.

The set of all k -permutations of $X = [n]$ is denoted by $P(n, k)$. In particular, $S_n = P(n, n)$.

Circular Permutations

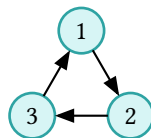
Definition 11: A *circular k -permutation* of $[n]$ arranges k distinct elements of $[n]$ in a circle. Two such arrangements are considered the same if one is a cyclic rotation of the other. The set of all circular k -permutations of $[n]$ is denoted $P_{c(n,k)}$.

Linear: all $3! = 6$ are distinct:



$(1, 2, 3), (1, 3, 2), (2, 1, 3),$
 $(2, 3, 1), (3, 1, 2), (3, 2, 1)$

Circular: rotations are identical:



$(1, 2, 3) \sim (2, 3, 1) \sim (3, 1, 2)$ – same circle

Only 2 distinct circles: $(1, 2, 3)$ and $(1, 3, 2)$

Counting Permutations

Theorem 1: For any natural numbers $0 \leq k \leq n$, we have

$$|P(n, k)| = n \cdot (n - 1) \cdot \dots \cdot (n - k + 1) = \frac{n!}{(n - k)!}$$

This formula is also called the *falling factorial* and denoted $n^{\underline{k}}$ or $(n)_k$.

Proof: A permutation is an injective map $\pi : [k] \rightarrow [n]$. We count the number of ways to pick such a map, picking the images one after the other. There are n ways to choose $\pi(1)$. Given a value for $\pi(1)$, there are $(n - 1)$ ways to choose $\pi(2)$ (since we may not choose $\pi(1)$ again). Continuing like this, there are $(n - i + 1)$ ways to pick $\pi(i)$, and the last value we pick is $\pi(k)$ with $(n - k + 1)$ possibilities.

Every k -permutation can be constructed like this in *exactly one way*. The total number of k -permutations is therefore given as the product:

$$|P(n, k)| = n \cdot (n - 1) \cdot \dots \cdot (n - k + 1) = \frac{n!}{(n - k)!}$$

□

Counting Circular Permutations

Theorem 2: For any natural numbers $0 \leq k \leq n$, we have

$$|P_c(n, k)| = \frac{n!}{k \cdot (n - k)!}$$

Proof: We doubly count $P(n, k)$:

1. $|P(n, k)| = \frac{n!}{(n-k)!}$ which we proved before.
2. $|P(n, k)| = |P_c(n, k)| \cdot k$ because every equivalence class in $P_c(n, k)$ contains k permutations from $P(n, k)$ since there are k ways to rotate a k -permutation.

From this we get $\frac{n!}{(n-k)!} = |P_c(n, k)| \cdot k$, which implies $|P_c(n, k)| = \frac{n!}{k \cdot (n - k)!}$. □

Unordered Arrangements

Definition 12: An *unordered arrangement* of k elements of X is a *multiset* $S = \langle X, r \rangle$ of size k .

In a multiset, X is the set of *types*, and for each type $x \in X$, r_x is its *repetition number*.

Example: Let $X = \{ \text{👉}, \text{🐼}, \text{🐱}, \text{🎴}, \text{🌵} \}$.

- An unordered arrangement of 7 elements could be $S = \{ \text{👉}, \text{👉}, \text{🐼}, \text{🐱}, \text{🐱}, \text{🐱}, \text{🌵} \}^*$.
- The same multiset could be written as $S = \{ 2 \text{👉}, 1 \text{🐼}, 3 \text{🐱}, 0 \text{🎴}, 1 \text{🌵} \}$.

Subsets

The most important special case of unordered arrangements is where all repetitions are 1, that is, $r_x = 1$ for all $x \in X$. Then S is simply a *subset* of X , denoted $S \subseteq X$.

Definition 13: A *k-combination* of X is an unordered arrangement of k *distinct* elements of X .

Note: The more standard term is *subset*. The term “combination” is only used to emphasize the selection process.

The set of all k -subsets of X is denoted $\binom{X}{k} = \{A \subseteq X \mid |A| = k\}$. If $|X| = n$, then

$$\binom{n}{k} := \left| \binom{X}{k} \right|$$

Example: The set of edges in a simple undirected graph consists of 2-subsets of its vertices: $E \subseteq \binom{V}{2}$.

Counting k -Combinations

Theorem 3: For $0 \leq k \leq n$, we have

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

Proof: $|P(n, k)| = \frac{n!}{(n-k)!} = \binom{n}{k} \cdot k!$

□

Four Counting Problems

	No repetition	With repetition
Ordered (k -perm., sequence)	$\frac{n!}{(n-k)!} = n^{\underline{k}}$	n^k
Unordered (k -comb., subset)	$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$	$\binom{k+n-1}{k}$

- n is the size of the ground set X .
- k is the size of the arrangement we want to count.
- “Ordered” means position matters: $(A, B) \neq (B, A)$.
- “Unordered” means only the selection matters: $\{A, B\} = \{B, A\}$.
- “With repetition” means the same element can appear multiple times.

Multisets

Multiset

Definition 14: A *multiset* is a modification of the concept of a set that allows for *repetitions* of its elements. Formally, it is denoted as a pair $M = \langle X, r \rangle$, where X is the *groundset* (the set of *types*) and $r : X \rightarrow \mathbb{N}_0$ is the *multiplicity function*.

Example: When the multiset is defined by enumeration, it is advisable to use the notation with the star:

$$M = \{a, b, a, a, b\}^* = \{3 \cdot a, 2 \cdot b\} \quad X = \{a, b\} \quad r_a = 3, r_b = 2$$

Example: Prime factorization of a natural number n is a multiset, e.g. $120 = 2^3 \cdot 3^1 \cdot 5^1$.

k -Combinations of a Multiset

Definition 15: Let X be a finite set of types, and let $M = \langle X, r \rangle$ be a finite multiset with repetition numbers $r_1, \dots, r_{|X|}$. A *k -combination of M* is a multiset $S = \langle X, s \rangle$ with types in X and repetition numbers $s_1, \dots, s_{|X|}$ such that $s_i \leq r_i$ for all $1 \leq i \leq |X|$, and $\sum_{i=1}^{|X|} s_i = k$.

Example: Consider $M = \{2 \text{ 🦊}, 1 \text{ 🥤}, 3 \text{ 🍉}, 1 \text{ 💎}\}$.

- $T = \{1 \text{ 🦊}, 2 \text{ 🍉}\}$ is a 3-combination of M .
- $T' = \{3 \text{ 💎}\}$ is not.

Counting k -combinations of a multiset is not as simple as it might seem...

k -Permutations of a Multiset

Definition 16: Let M be a finite multiset with set of types X . A *k -permutation of M* is an ordered arrangement of k elements of M where different orderings of elements of the same type are *not distinguished*. This is an ordered multiset with types in X and repetition numbers $s_1, \dots, s_{|X|}$ such that $s_i \leq r_i$ for all $1 \leq i \leq |X|$, and $\sum_{i=1}^{|X|} s_i = k$.

Note: There might be several elements of the same type compared to a permutation of a set (where each repetition number equals 1).

Example: Let $M = \{2 \text{ 🦊}, 1 \text{ 🥤}, 3 \text{ 🍉}, 1 \text{ 💎}\}$, then $T = (\text{💎}, \text{🍉}, \text{🍉}, \text{🦊})$ is a 4-permutation of multiset M .

Binomial Theorem

Theorem 4: The expansion of any non-negative integer power n of the binomial $(x + y)$ is a sum

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} \cdot x^k \cdot y^{n-k}$$

where each $\binom{n}{k}$ is a positive integer known as a *binomial coefficient*, defined as

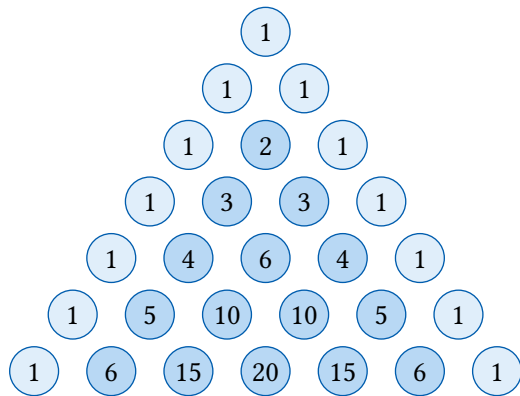
$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!} = \frac{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1)}{k \cdot (k-1) \cdot (k-2) \cdot \dots \cdot 2 \cdot 1}$$

Pascal's Triangle

Row n of *Pascal's triangle* lists $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$. Each entry is the sum of the two entries above it: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Key identities:

- **Row sum:** $\sum_{k=0}^n \binom{n}{k} = 2^n$ (set $x = y = 1$)
- **Symmetry:** $\binom{n}{k} = \binom{n}{n-k}$
- **Absorption:** $n \binom{n-1}{k-1} = (k+1) \binom{n}{k+1}$
- **Vandermonde:** $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$



Multinomial Theorem

Theorem 5: The generalization of the binomial theorem:

$$(x_1 + \dots + x_r)^n = \sum_{\substack{0 \leq k_1, \dots, k_r \leq n \\ k_1 + \dots + k_r = n}} \binom{n}{k_1, \dots, k_r} \cdot x_1^{k_1} \cdot \dots \cdot x_r^{k_r}$$

Multinomial coefficients are defined as

$$\binom{n}{k_1, \dots, k_r} = \frac{n!}{k_1! \cdot \dots \cdot k_r!}$$

Note: Binomial coefficients are special cases of multinomial coefficients ($r = 2$):

$$\binom{n}{k} = \binom{n}{k_1, k_2} = \binom{n}{k, n-k} = \frac{n!}{k! \cdot (n-k)!}$$

Multinomial Theorem [2]

Proof: Expand $(x_1 + \dots + x_r)^n$ by choosing one term from each of the n factors. A monomial $x_1^{k_1} \cdot \dots \cdot x_r^{k_r}$ with $k_1 + \dots + k_r = n$ is produced whenever we choose x_i from exactly k_i of the n factors. The number of ways to assign the n factor-positions to the r variables in groups of sizes $k_1, \dots, k_r$ is

$$\binom{n}{k_1, \dots, k_r} = \frac{n!}{k_1! \cdot \dots \cdot k_r!}.$$

□

Theorem 6: Let S be a finite multiset with k different types and repetition numbers $r_1, \dots, r_k$. Let the size of S be $n = r_1 + \dots + r_k$. Then the number of n -permutations of S equals

$$\binom{n}{r_1, \dots, r_k}$$

Proof: In an n -permutation there are n positions that need to be assigned a type.

First, choose the r_1 positions for the first type, there are $\binom{n}{r_1}$ ways to do so. Then, assign r_2 positions for the second type, out of the $(n - r_1)$ positions that are still available, there are $\binom{n-r_1}{r_2}$ ways to do so. Continue for all k types. The total number of choices will be:

Multinomial Theorem [3]

$$\binom{n}{r_1} \cdot \binom{n-r_1}{r_2} \cdot \dots \cdot \binom{n-r_1-r_2-\dots-r_{k-1}}{r_k} = \binom{n}{r_1, \dots, r_k}$$



k -Combinations of an *Infinite Multiset*

Example: Suppose you have a *sufficiently large* amount of each type of fruit (🍌, 🍎, 🍐) in the supermarket, and you want to buy *two* fruits. How many choices do you have?

There are exactly *six* combinations: {🍌, 🍌}, {🍌, 🍎}, {🍌, 🍐}, {🍎, 🍎}, {🍎, 🍐}, {🍐, 🍐}.

Note that your selection is *not ordered*, so {🍐, 🍎} and {🍎, 🍐} are considered the *same* choice.

k -Combinations of an *Infinite Multiset* [2]

Theorem 7: Let $k, s \in \mathbb{N}$ and let S be a multiset with s types and large repetition numbers (each $r_1, \dots, r_s$ is *at least k*), then the number of k -combinations of S equals

$$\binom{k+s-1}{k} = \binom{k+s-1}{s-1}$$

Proof: Let $S = \langle X, r_\infty \rangle = \{\infty \text{ 🍌}, \infty \text{ 🍎}, \infty \text{ 🍇}\}$ with $r_x = \infty$ and $|X| = s = 3$.

- Let $k = 5$ (as an example). Consider a 5-combination of S : { 🍌 🍌 🍌 🍌 🍌, 🍎, 🍌, 🍇, 🍇 }.
- Reorder and group: { 🍌 🍌 🍌 | 🍎 | 🍇 🍇 }.
- Convert to *dots* and *bars*: ••• | • | ••
- Represent as a 2-type multiset: $M = \{k \cdot \bullet, (s-1) \cdot |\}$
- Observe: each *permutation* of k dots and $(s-1)$ bars corresponds *uniquely* to a *k -combination* of S .
- Permute the 2-type multiset: $\binom{k+s-1}{k, s-1}$ ways, by Theorem 5.

This method is also known as *Stars and Bars*. □

Compositions

Weak Compositions

Definition 17: A *weak composition* of a non-negative integer $k \geq 0$ into s parts is a *solution* to the equation $b_1 + \dots + b_s = k$, where each $b_i \geq 0$.

Example: Let $k = 5$, $s = 3$. Possible non-negative integer solutions for $b_1 + b_2 + b_3 = 5$ are:

- $(b_1, b_2, b_3) = (1, 1, 3)$
- $(b_1, b_2, b_3) = (1, 3, 1)$
- $(b_1, b_2, b_3) = (2, 0, 3)$
- $(b_1, b_2, b_3) = (0, 5, 0)$
- ... (total 21 solutions)

Note: If M is a multiset over groundset $\{1, \dots, s\}$ with all multiplicities infinite ($r_i = \infty$), then for $k \geq 0$, the number of sub-multisets of M of size k is exactly the number of weak compositions of k into s parts.

Counting Weak Compositions

Theorem 8: There are $\binom{k+s-1}{k, s-1}$ *weak compositions* of $k > 0$ into s parts.

Proof: Observe that $k = \overbrace{1+1+\dots+1+1}^{k \text{ ones}}$
 $\qquad\qquad\qquad \underbrace{\qquad}_{b_1} \quad \underbrace{\qquad}_{b_i} \quad \underbrace{\qquad}_{b_s}$

Use the *stars-and-bars* method to count the number of s groups composed of k “ones”. □

Example: Let $k = 3$. There are $\binom{3+3-1}{3, 3-1} = \binom{5}{3} = \binom{5}{2} = 10$ ways to decompose $k = 3$ into $s = 3$ parts:

$$\begin{aligned} k = 3 &= \\ &= 0 + 1 + 2 = 0 + 2 + 1 \\ &= 1 + 0 + 2 = 1 + 2 + 0 = 1 + 1 + 1 \\ &= 2 + 0 + 1 = 2 + 1 + 0 \\ &= 3 + 0 + 0 = 0 + 3 + 0 = 0 + 0 + 3 \end{aligned}$$

Compositions

Definition 18: A *composition* of a positive integer $k \geq 1$ into s *positive* parts is a *solution* to the equation $b_1 + \dots + b_s = k$, where each $b_i > 0$.

Theorem 9: There are $\binom{k-1}{s-1}$ *compositions* of $k > 0$ into s positive parts.

Proof: Line up k ones in a row. A composition $b_1 + \dots + b_s = k$ with each $b_i \geq 1$ corresponds to choosing $s - 1$ of the $k - 1$ *gaps between consecutive ones* as dividers. There are $\binom{k-1}{s-1}$ such choices. $\square$

Compositions [2]

Theorem 10: The total number of compositions of $k > 0$ into *some* number of positive parts is

$$\sum_{s=1}^k \binom{k-1}{s-1} = 2^{k-1}$$

Proof: Substituting $j = s - 1$:

$$\sum_{s=1}^k \binom{k-1}{s-1} = \sum_{j=0}^{k-1} \binom{k-1}{j} = (1+1)^{k-1} = 2^{k-1}$$

by the Binomial Theorem applied to $(1+1)^{k-1}$. □

Parallel Summation Identity

Q: How many integer solutions are there to the *inequality* $b_1 + \dots + b_s \leq k$, where each $b_i \geq 0$?

Theorem 11:
$$\sum_{m=0}^k \binom{m+s-1}{m} = \binom{k+s}{k}$$

Proof (*hint*): Introduce a “dummy” variable b_{s+1} to take up the *slack* between $b_1 + \dots + b_s$ and k . Construct a bijection with the solutions to $b_1 + \dots + b_s + b_{s+1} = k$, where $b_i \geq 0$. □

Set Partitions

Set Partitions

Definition 19: A *partition* of a set X is a set of non-empty subsets of X such that every element of X belongs to exactly one of these subsets.

Equivalently, a family of sets P is a partition of X iff:

1. The family P does not contain the empty set: $\emptyset \notin P$.
2. The union of P is X , that is, $\bigcup_{A \in P} A = X$. The sets in P are said to *cover* X .
3. The intersection of any two distinct sets in P is empty: $\forall A, B \in P. (A \neq B) \rightarrow (A \cap B = \emptyset)$.
The sets in P are said to be *pairwise disjoint* or *mutually exclusive*.

The sets in P are called *blocks*, *parts*, or *cells*, of the partition.

The block in P containing an element $x \in X$ is denoted by $[x]$.

Examples of Set Partitions

Example: The empty set $X = \emptyset$ has exactly one partition, $P = \emptyset$.

Example: Any singleton set $X = \{x\}$ has exactly one partition, $P = \{\{x\}\}$.

Example: For any non-empty proper subset $A \subset U$, the set A and its complement form a partition of U , namely $P = \{A, U - A\}$.

Example: The set $X = \{1, 2, 3\}$ has five partitions:

1. $\{\{1\}, \{2\}, \{3\}\}$ or $1 \mid 2 \mid 3$
2. $\{\{1\}, \{2, 3\}\}$ or $1 \mid 2 \ 3$
3. $\{\{1, 2\}, \{3\}\}$ or $1 \ 2 \mid 3$
4. $\{\{1, 3\}, \{2\}\}$ or $1 \ 3 \mid 2$
5. $\{\{1, 2, 3\}\}$ or $1 \ 2 \ 3$

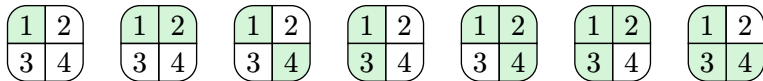
Example: The following are *not* partitions of $\{1, 2, 3\}$:

- $\{\{\}, \{1, 3\}, \{2\}\}$, because it contains the empty set.
- $\{\{1, 2\}, \{2, 3\}\}$, because the element 2 is contained in more than one block.
- $\{\{1\}, \{3\}\}$, because no block contains the element 3.

Counting Set Partitions

Definition 20: The number of partitions of a set X (of size $n = |X|$) into k non-empty blocks (“unlabeled subsets”) is called a *Stirling number of the second kind* and denoted $S(n, k)$ or $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$.

Example: Let $X = \{1, 2, 3, 4\}$, $k = 2$. There are 7 possible partitions:



Theorem 12: Let $\left\{ \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\} = 0$ for $n \geq 1$, $\left\{ \begin{smallmatrix} 0 \\ k \end{smallmatrix} \right\} = 0$ for $k \geq 1$, and $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$. For $n, k \geq 1$, we have:

$$\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\} + k \cdot \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\}$$

Proof (informal): Consider the element n in any partition of $[n]$ into k blocks. There are exactly two cases:

Counting Set Partitions [2]

- $\{n\}$ forms a singleton block on its own. The remaining $n - 1$ elements are partitioned into $k - 1$ non-empty blocks: $\left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\}$ ways.
- n joins an existing block of a partition of $[n - 1]$. Start with any partition of $[n - 1]$ into k blocks ($\left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\}$ ways), then insert n into one of the k blocks: $k \cdot \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\}$ ways.

These two cases are exhaustive and disjoint.



Bell Numbers

Definition 21: The total number of partitions of a set X of size $n = |X|$ (into an arbitrary number of non-empty blocks) is called a *Bell number* and denoted B_n .

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

Note: Consider the special case of $n = 0$. There is exactly *one* partition of $\emptyset$ into non-empty parts: $\emptyset = \bigcup_{A \in \emptyset} A \in \emptyset$. Every $A \in \emptyset$ is non-empty, since no such A exists. Thus, we have $B_0 = S(0, 0) = 1$.

Bell Numbers [2]

Theorem 13: For $n \geq 1$, we have the recursive identity for Bell numbers:

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k$$

Proof: Every partition of $[n]$ has one part that contains the number n . In addition to n , this part also contains k other numbers (for some $0 \leq k \leq n-1$). The remaining $n-1-k$ elements are partitioned arbitrarily. From this correspondence, we obtain the desired identity:

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_{n-1-k} = \sum_{k=0}^{n-1} \binom{n-1}{n-1-k} B_{n-1-k} = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k$$

□

Integer Partitions

Integer Partitions

The integer 4 can be written as a sum of positive integers in exactly 5 ways: $4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1$. Order does *not* matter: $3 + 1$ and $1 + 3$ are the same partition.

Definition 22: An *integer partition* of a positive integer $n \geq 1$ into k *positive* parts is a solution to the equation $n = a_1 + \dots + a_k$, where $a_1 \geq a_2 \geq \dots \geq a_k \geq 1$.

- The number of such partitions is denoted $p_k(n)$ and defined recursively:

$$p_k(n) = \begin{cases} 0 & \text{if } k > n \\ 0 & \text{if } n \geq 1 \text{ and } k = 0 \\ 1 & \text{if } n = k = 0 \\ p_k(n - k) + p_{k-1}(n - 1) & \text{if } 1 \leq k \leq n \end{cases}$$

- The *partition function* $p(n)$ counts all partitions of n :

$$p(n) = \sum_{k=0}^n p_k(n)$$

Integer Partitions [2]

Rapid growth of $p(n)$: $p(5) = 7$, $p(10) = 42$, $p(50) = 204{,}226$, $p(100) = 190{,}569{,}292$. Hardy and Ramanujan (1918) proved:

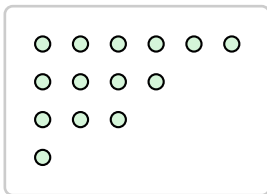
$$p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{2n/3}}$$

so the number of partitions grows *sub-exponentially* but faster than any polynomial.

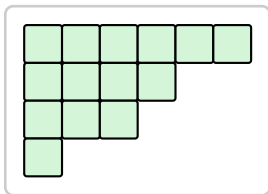
Ferrers Diagrams and Young Tableaux

Example: Consider an integer partition: $14 = 6 + 4 + 3 + 1$.

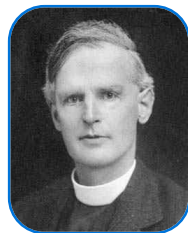
Ferrer Diagram



Young Tableaux



Norman Ferrer



Alfred Young

Inclusion–Exclusion

“There are very few things which we know, which are not capable of being reduced to a mathematical reasoning.”

— Abraham de Moivre, The Doctrine of Chances (1718)

Motivating Example

Example: How many integers in $\{1, \dots, 100\}$ are divisible by 2 *or* by 3?

- Let $A = \{k \in [100] \mid 2 \mid k\}$, so $|A| = 50$.
- Let $B = \{k \in [100] \mid 3 \mid k\}$, so $|B| = 33$.
- Their intersection $A \cap B = \{k \in [100] \mid 6 \mid k\}$ has $|A \cap B| = 16$.

By inclusion-exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B| = 50 + 33 - 16 = 67$$

Generalizing this to an arbitrary number of sets gives the full PIE theorem.

Principle of Inclusion–Exclusion (PIE)

Theorem 14: Let X be a finite set and $P_1, \dots, P_m$ *properties*.

- Define $X_i = \{x \in X \mid x \text{ has } P_i\}$, i.e. the set of all elements from X having a property P_i .
- Define for $S \subseteq [m]$ the set $N(S) = \{x \in X \mid \forall i \in S : x \text{ has } P_i\}$. Observe: $N(S) = \bigcap_{i \in S} X_i$.

The number of elements of X that satisfy *none* of the properties $P_1, \dots, P_m$ is given by

$$|X \setminus (X_1 \cup \dots \cup X_m)| = \sum_{S \subseteq [m]} (-1)^{|S|} |N(S)| \quad (1)$$

Proof: Consider any $x \in X$. If $x \in X$ has none of the properties, then $x \in N(\emptyset)$ and $x \notin N(S)$ for any other $S \neq \emptyset$. Hence x *contributes 1* to the sum (1).

If $x \in X$ has exactly $k \geq 1$ of the properties, call this set $T \in \binom{[m]}{k}$. Then $x \in N(S)$ iff $S \subseteq T$.

The *contribution of x* to the sum (1) is $\sum_{S \subseteq T} (-1)^{|S|} = \sum_{i=0}^k \binom{k}{i} (-1)^i = 0$, i.e. *zero*. □

Note: In the last step, we used that for any $y \in \mathbb{R}$ we have $(1 - y)^k = \sum_{i=0}^k \binom{k}{i} (-y)^i$ which implies (for $y = 1$) that $0 = \sum_{i=0}^k \binom{k}{i} (-1)^i$.

Counting the Union

Corollary 14.1:

$$\left| \bigcup_{i \in [m]} X_i \right| = \sum_{\emptyset \neq S \subseteq [m]} (-1)^{|S|-1} |N(S)|$$

Applications of PIE

Step 1: Define “bad” properties

Identify elements of a universe X to *exclude*: those having *at least one* of properties $P_1, \dots, P_m$.

Target: count $X \setminus (X_1 \cup \dots \cup X_m)$.

Step 2: Count $N(S)$

For each $S \subseteq [m]$, determine $N(S)$ – number of elements of X having *all* bad properties P_i for $i \in S$.

Step 3: Apply PIE

Use Theorem 14 to compute a closed formula:

$$|X \setminus (X_1 \cup \dots \cup X_m)| = \sum_{S \subseteq [m]} (-1)^{|S|} N(S)$$

Counting Surjections via PIE

Theorem 15: The number of surjections from $[k]$ to $[n]$ is given by

$$\left| \left\{ f : [k] \xrightarrow{\text{surj.}} [n] \right\} \right| = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^k$$

Proof: Let X be the set of all maps from $[k]$ to $[n]$.

1. *Define bad properties:* Define the “bad” property P_i for $i \in [n]$ as “ i is not in the image of f ”, i.e.

$$f : [k] \rightarrow [n] \text{ has property } P_i \iff \forall j \in [k] : f(j) \neq i$$

The *surjective* functions are exactly those functions that *do not* have any of the “bad” properties.

2. *Count $N(S)$:* We claim $N(S) = (n - |S|)^k$ for any $S \subseteq [n]$. To see this, observe that f has all properties with indices from S if and only if $f(i) \notin S$ for all $i \in [k]$. In other words, f must be a function from $[k]$ to $[n] \setminus S$, and there are $(n - |S|)^k$ of those.

Counting Surjections via PIE [2]

3. *Apply PIE:* Using Theorem 14, the number of surjections from $[k]$ to $[n]$ is

$$\begin{aligned} |X \setminus (X_1 \cup \dots \cup X_n)| &\stackrel{\text{PIE}}{=} \sum_{S \subseteq [n]} (-1)^{|S|} |N(S)| \\ &= \sum_{S \subseteq [n]} (-1)^{|S|} (n - |S|)^k \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i} (n - i)^k \end{aligned}$$

In the last step, we used that $(-1)^{|S|} (n - |S|)^k$ only depends on the size of S , and there are $\binom{n}{i}$ sets $S \subseteq [n]$ of size i .

□

Useful Corollaries

Corollary 15.1: Consider the case $n = k$. A function from $[n]$ to $[n]$ is a *surjection* iff it is a *bijection*. Since there are $n!$ bijections on $[n]$ (namely, all permutations), we have the following identity:

$$n! = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^n$$

Corollary 15.2: A surjection from $[k]$ to $[n]$ can be seen as a partition of $[k]$ into n non-empty distinguishable (labeled) parts (the map assigns a part to each $i \in [n]$).

Since the partition of $[k]$ into n non-empty indistinguishable parts is denoted $s_n^{\text{II}}(k)$, and there are $n!$ ways to assign labels to n parts, we obtain that the number of surjections is equal to $n!s_n^{\text{II}}(k)$:

$$n!s_n^{\text{II}}(k) = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^k$$

Derangements

Theorem 16: The *derangements* D_n on n elements are permutations of $[n]$ without fixed points.

The number of derangements is given by

$$|D_n| = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)!$$

Proof: Let X be the set of all permutations of $[n]$.

1. Define the “bad” property P_i to mean “ π has a fixpoint i ” ($i \in [n]$):

$$\pi \in X \text{ has property } P_i \iff \pi(i) = i$$

2. We claim $N(S) = (n - |S|)!$ for any $S \subseteq [n]$.

Indeed, $\pi \in X$ has all properties with indices from S if and only if all $i \in S$ are fixed points of π . On the other elements, i.e. on $[n] \setminus S$, π may be an arbitrary bijection, so there are $(n - |S|)!$ choices for π .

Derangements [2]

3. Using Theorem 14, the number of derangements is given by

$$\begin{aligned} |X \setminus (X_1 \cup \dots \cup X_n)| &\stackrel{\text{PIE}}{=} \sum_{S \subseteq [n]} (-1)^{|S|} |N(S)| \\ &= \sum_{S \subseteq [n]} (-1)^{|S|} (n - |S|)! \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i} (n - i)! \end{aligned}$$

In the last step, we used that $(-1)^{|S|} (n - |S|)!$ only depends on the size of S , and there are $\binom{n}{i}$ sets $S \subseteq [n]$ of size i .

□

Derangements [3]

Hat-check problem: n guests check their hats; the attendant loses the tickets. As $n \rightarrow \infty$, the probability that no guest gets their own hat back converges to

$$\lim_{n \rightarrow \infty} \frac{|D_n|}{n!} = \sum_{i=0}^{\infty} \frac{(-1)^i}{i!} = \frac{1}{e} \approx 36.8\%$$

This probability barely changes for $n \geq 5$.

Generating Functions

“A generating function is a clothesline on which we hang up a sequence of numbers for display.”

— Herbert Wilf, generatingfunctionology

Generating Functions

A generating function is a device somewhat similar to a bag. Instead of carrying many little objects detachedly, which could be embarrassing, we put them all in a bag, and then we have only one object to carry, the bag.

— George Pólya, Mathematics and Plausible Reasoning [1]

Encode a sequence (a_n) as coefficients of a formal power series $G(x) = \sum_{n=0}^{\infty} a_n x^n$, then manipulate $G(x)$ *algebraically* (multiply, differentiate, compose) to derive counting identities and closed forms.

Counting with Polynomials

Problem. Two fair dice. How many outcomes give sum n ?

Encode one die as a polynomial: the term x^k stands for “rolling k is possible.”

$$D(x) = x + x^2 + x^3 + x^4 + x^5 + x^6$$

Two independent dice $\rightarrow$ multiply:

$$D(x)^2 = x^2 + 2x^3 + 3x^4 + 4x^5 + 5x^6 + \underline{6x^7} + 5x^8 + 4x^9 + 3x^{10} + 2x^{11} + x^{12}$$

The coefficient of x^n gives the count for sum n . For sum 7: $[x^7]D(x)^2 = 6$. ✓

One multiplication solves the problem for *every* n simultaneously.

This polynomial is a *generating function* — now let's understand what they are and why this works.

Ordinary Generating Functions

Definition 23: An *ordinary generating function* (OGF) of a sequence a_n is a *power series*

$$G(a_n; x) = \sum_{n=0}^{\infty} a_n x^n$$

Example: The *sequence* $a_n = (a_0, a_1, a_2, \dots)$ is *generated* by the OGF $G(x) = a_0 + a_1x + a_2x^2 + \dots$

Example: $G(x) = 3 + 8x^2 + x^3 + \frac{1}{7}x^5 + 100x^6 + \dots$ *generates* the sequence $(3, 0, 8, 1, 0, \frac{1}{7}, 100, 0, \dots)$

Example: Consider a long division of 1 by $(1 - x)$, the result is an infinite power series

$$\frac{1}{1-x} = 1 + x^1 + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n$$

Note that all coefficients are 1. Thus, the generating function of $(1, 1, 1, \dots)$ is $G(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$.

The Core Generating Function

Another proof that $(1, 1, 1, \dots)$ is generated by $G(x) = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = S$:

$$S = 1 + x + x^2 + x^3 + \dots$$

$$x \cdot S = x + x^2 + x^3 + \dots$$

$$S - x \cdot S = 1$$

$$\text{Thus, } S = \frac{1}{1-x}.$$

The generating function $G(x) = 1 + x + x^2 + \dots$ is also known as the *Maclaurin series* of $\frac{1}{1-x}$.

More Examples of Generating Functions

Formula	Power series	Sequence	Description
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$	$(1, 1, 1, \dots)$	constant 1
$\frac{2}{1-x}$	$\sum_{n=0}^{\infty} 2x^n = 2 + 2x + 2x^2 + 2x^3 + \dots$	$(2, 2, 2, \dots)$	constant 2
$\frac{x}{1-x}$	$\sum_{n=1}^{\infty} x^n = 0 + x + x^2 + x^3 + \dots$	$(0, 1, 1, 1, \dots)$	right shift
$\frac{1}{1+x}$	$\sum_{n=0}^{\infty} (-1)^n x^n = 0 + 1 - x + x^2 - x^3 + \dots$	$(1, -1, 1, \dots)$	sign-alternating 1's
$\frac{1}{1-3x}$	$\sum_{n=0}^{\infty} 3^n x^n = 1 + 3x + 9x^2 + 27x^3 + \dots$	$(1, 3, 9, \dots)$	powers of 3

More Examples of Generating Functions [2]

Formula	Power series	Sequence	Description
$\frac{1}{1-x^2}$	$\sum_{n=0}^{\infty} x^{2n} = 1 + x^2 + x^4 + x^6 + \dots$	$(1, 0, 1, 0, \dots)$	regular gaps
$\frac{1}{(1-x)^2}$	$\sum_{n=0}^{\infty} (n+1)x^n = 1 + 2x + 3x^2 + 4x^3 + \dots$	$(1, 2, 3, 4, \dots)$	natural numbers

$$\begin{aligned}
 \frac{1-x^{n+1}}{1-x} &= \frac{1}{1-x} - \frac{x^{n+1}}{1-x} = \\
 &\triangleq (1, 1, 1, \dots) - \underbrace{(0, 0, \dots, 0, 1, 1, \dots)}_{n+1 \text{ zeros}} = \\
 &= \underbrace{(1, 1, \dots, 1, 0, 0, \dots)}_{n+1 \text{ ones}} = \\
 &\triangleq 1 + x + x^2 + \dots + x^n
 \end{aligned}$$

Exercises

Example: Find the GF for odd numbers: $(1, 3, 5, 7, \dots)$.

Hint: write $A(x) = 1 + 3x + 5x^2 + 7x^3 + \dots$, then compute $x \cdot A(x)$ and subtract from $A(x)$.

What familiar series do you see in the result?

Example: Find the GF for $(1, 3, 7, 15, 31, 63, \dots)$, which satisfies $a_n = 3a_{n-1} - 2a_{n-2}$ with $a_0 = 1, a_1 = 3$.

Hint: write $A(x)$, $-3xA(x)$, and $+2x^2A(x)$ stacked and add them column by column.

Most terms cancel because of the recurrence. What remains?

Solutions

Example: Odd numbers. Let $A(x) = 1 + 3x + 5x^2 + 7x^3 + \dots$

$$\begin{array}{r} A(x) = 1 + 3x + 5x^2 + 7x^3 + 9x^4 + \dots \\ x \cdot A(x) = + x + 3x^2 + 5x^3 + 7x^4 + \dots \\ \hline A - x \cdot A = 1 + 2x + 2x^2 + 2x^3 + 2x^4 + \dots \end{array}$$

The result is $1 + 2(x + x^2 + x^3 + \dots) = 1 + 2 \cdot \frac{x}{1-x}$, so:

$$A(x) = \frac{1 + 2 \cdot \frac{x}{1-x}}{1-x} = \frac{1-x+2x}{(1-x)^2} = \frac{1+x}{(1-x)^2}$$

Check: $\frac{1+x}{(1-x)^2} = (1, 1, 1, \dots) + (0, 2, 4, 6, 8, \dots) = (1, 3, 5, 7, 9, \dots)$. ✓

Solutions [2]

Example: Recurrence sequence. Stack A , $-3x \cdot A$, $+2x^2 \cdot A$ and add:

$$\begin{array}{rcl} A(x) & = & 1 + 3x + 7x^2 + 15x^3 + 31x^4 + \dots \\ -3x \cdot A(x) & = & -3x - 9x^2 - 21x^3 - 45x^4 - \dots \\ 2x^2 \cdot A(x) & = & 2x^2 + 6x^3 + 18x^4 + \dots \\ \hline (1 - 3x + 2x^2) \cdot A(x) & = & 1 \end{array}$$

Every column beyond x^0 vanishes because $a_n - 3a_{n-1} + 2a_{n-2} = 0$ by the recurrence.

$$\text{Thus } A(x) = \frac{1}{1 - 3x + 2x^2}.$$

Solving Combinatorial Problems via Generating Functions

Example: Find the number of integer solutions to $y_1 + y_2 + y_3 = 12$ with $0 \leq x_i \leq 6$.

- Possible values for y_1 are $0 \leq y_1 \leq 6$.
 - ▶ There is a *single* way to select $y_1 = 0$. The same for other values among $1, \dots, 6$.
 - ▶ There are *no* ways to select any value of y_1 higher than 6.
 - ▶ The *number of ways to select y_1 to be equal to n* forms a sequence $(1, 1, 1, 1, 1, 1, 0, \dots)$.
 - ▶ Write this sequence as a polynomial $x^0 + x^1 + \dots + x^6$.
 - ▶ Do the same for y_2 and y_3 (*in isolation!*).
- Since all combinations of y_1, y_2 and y_3 are valid non-conflicting solutions, we can multiply those polynomials and obtain the *generating function* $G(x) = (1 + x + x^2 + \dots + x^6)^3$.
 - ▶ For each n , the coefficient of x^n in $G(x)$ is the number of integer solutions to $x_1 + x_2 + x_3 = n$.
 - ▶ In particular, we are interested in the coefficient of x^{12} in $G(x)$, denoted $[x^{12}]G(x)$.
 - ▶ Use ~~pen and paper~~ Wolfram Alpha to expand $G(x)$:

$$G(x) = x^{18} + 3x^{17} + 6x^{16} + \dots + \underline{28x^{12}} + \dots + 6x^2 + 3x + 1$$

- The *answer* is $[x^{12}]G(x) = 28$ solutions.

Slightly More Complex Combinatorial Problem

Example: Suppose we have marbles of three different colors (●, ●, ●), and we want to *count* the number of ways to select 20 marbles, such that:

- There are an even number of ●: $1 + x^2 + x^4 + \dots + x^{20}$.
- There are at least 12 ●: $x^{12} + x^{13} + \dots + x^{20}$.
- There are at most 5 ●: $1 + x + x^2 + x^3 + x^4 + x^5$.

Multiply polynomials and find $[x^{20}]G(x)$:

$$\begin{aligned} & [x^{20}](1 + x^2 + x^4 + \dots + x^{20})(x^{12} + x^{13} + \dots + x^{20})(1 + x + x^2 + x^3 + x^4 + x^5) = \\ & = [x^{20}](x^{45} + 2x^{44} + \dots + \underline{21x^{20}} + \dots + 2x^{13} + x^{12}) \\ & = 21 \end{aligned}$$

Using Power Series in Combinatorial Problems

Example: Find the number of integer solutions to $a_1 + a_2 + a_3 = 12$ with $a_1 \geq 2, 3 \leq a_2 \leq 6, 0 \leq a_3 \leq 9$.

- Compose the generating function:

$$G(x) = (x^2 + x^3 + \dots) \cdot (x^3 + x^4 + x^5 + x^6) \cdot (1 + x + x^2 + \dots + x^9)$$

- Substitute the power series with the corresponding simple forms:

$$G(x) = \left(x^2 \cdot \frac{1}{1-x}\right) \cdot \left(x^3 \cdot \frac{1-x^4}{1-x}\right) \cdot \left(\frac{1-x^{10}}{1-x}\right)$$

- Expand the series:

$$G(x) = x^5 + 3x^6 + 6x^7 + 10x^8 + 14x^9 + 18x^{10} + 22x^{11} + \underline{26x^{12}} + 30x^{13} + \\ 34x^{14} + 37x^{15} + 39x^{16} + 40x^{17} + \dots + 40x^n + \dots$$

- Sequence: $(g_n) = (0, 0, 0, 0, 0, 1, 3, 6, 10, 14, 18, 22, 26, 30, 34, 37, 39, \overline{40}, \dots)$
- Answer for $n = 12$ is $[x^{12}]G(x) = 26$.

Operations on Generating Functions

Let $F(x) = \sum_{n=0}^{\infty} a_n x^n$ and $G(x) = \sum_{n=0}^{\infty} b_n x^n$ be ordinary generating functions.

Operation	Result
Differentiate $F(x)$ term-wise	$F'(x) = \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n$
Multiply $F(x)$ by a scalar $\lambda \in \mathbb{R}$ term-wise	$\lambda F(x) = \sum_{n=0}^{\infty} \lambda a_n x^n$
Add $F(x)$ and $G(x)$ term-wise	$F(x) + G(x) = \sum_{n=0}^{\infty} (a_n + b_n)x^n$
Multiply $F(x)$ and $G(x)$ term-wise (<i>Cauchy product</i> , or <i>convolution</i>)	$F(x) \cdot G(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n$

Catalan Numbers via Generating Functions

So far, we *composed* generating functions from known polynomials and read off coefficients.

Now we turn the idea around: start from a *recursive counting argument*, derive a *functional equation* for $F(x)$, and solve it.

This requires a heavier tool — *Newton's binomial theorem* for real-valued exponents — but the payoff is one of the most famous sequences in combinatorics.

Historical note: Eugène Catalan (1814–1894) studied these numbers in the context of polygon triangulation. They appear in dozens of seemingly unrelated counting problems — a phenomenon that generating functions help explain.

Well-Formed Parenthesis Expressions

Example: Find the number of *well-formed parenthesis expressions* with n pairs of parenthesis.

For example, “(() ())” is a well-formed parenthesis expression with 4 pairs of parenthesis.

Formally, a permutation of the multiset $\{n \cdot "(", n \cdot ")"\}$ is *well-formed* if reading it from left to right and counting “+1” for every opening parenthesis “(“ and “−1” for every closing parenthesis “)” never yields a negative number at any time.

Every well-formed expression with $n \geq 1$ pairs of parenthesis starts with “(“ and there is a unique matching “)” such that the sequence in between and the sequence after are well-formed. For example:

$$()()() \quad (())() \quad ()(())()$$

In other words, a well-formed expression with n pairs of parenthesis is obtained by putting a well-formed expression with k pairs in between “(“ and “)” and then appending a well-formed expression with $n - k - 1$ pairs of parenthesis. This gives the equation:

$$a_n = \sum_{k=0}^{n-1} a_k a_{n-k-1}$$

Well-Formed Parenthesis Expressions [2]

Let $F(x)$ be a generating function for a_n , then we know:

$$\begin{aligned} F(x) &= \sum_{n=0}^{\infty} a_n x^n = 1 + \sum_{n=1}^{\infty} \left(\sum_{k=0}^{n-1} a_k a_{n-k-1} \right) x^n = 1 + \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k a_{n-k} \right) x^{n+1} \\ &= 1 + x \cdot \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k a_{n-k} \right) x^n = 1 + x \cdot F(x)^2 \end{aligned}$$

Solving the equation $xF^2 - F + 1 = 0$ for F gives:

$$F(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$$

We are only interested in the root with the *minus* sign, since $\lim_{x \rightarrow 0} F(x) = a_0 = 1$:

$$\lim_{x \rightarrow 0} F(x) = \lim_{x \rightarrow 0} \frac{1 - \sqrt{1-4x}}{2x} = \lim_{x \rightarrow 0} \frac{\frac{2}{\sqrt{1-4x}}}{2} = 1$$

Newton's Binomial Theorem

To extract the coefficients of $F(x) = \frac{1-\sqrt{1-4x}}{2x}$, we need to expand $\sqrt{1-4x}$ as a power series.

This requires extending the binomial theorem to *real* exponents.

Let's revisit the binomial theorem:

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = \sum_{k=0}^{\infty} \binom{n}{k} x^k \quad \forall n \in \mathbb{N}$$

where $\binom{n}{k} = 0$ for $k > n$.

Note: This shows that $(1+x)^n$ is the generating function for the series $(a_k)_{k \in \mathbb{N}}$ with $a_k = \binom{n}{k}$.

We can extend this result from natural numbers $n \in \mathbb{N}$ to any *real* number $n \in \mathbb{R}$.

Binomial Coefficients for Real Numbers

Definition 24: Let $p(n, k) = n \cdot (n - 1) \cdot \dots \cdot (n - k + 1)$, also called the *falling factorial* $n^{\underline{k}}$.

Extend the definition of binomial coefficients for real numbers $n, k \in \mathbb{R}$:

$$\binom{n}{k} = \frac{p(n, k)}{k!}$$

Note: This definition aligns with the definition of binomial coefficients for natural numbers:

$$\binom{n}{k} = \frac{n!}{k! \cdot (n - k)!} = \frac{n \cdot (n - 1) \cdot \dots \cdot (n - k + 1)}{k!}$$

Example: Consider the number “ $-7/2$ choose 5”:

$$\binom{-7/2}{5} = \frac{-7/2 \cdot -9/2 \cdot -11/2 \cdot -13/2 \cdot -15/2}{5!} = -\frac{9009}{256}$$

Note: $p(n, 0) = 1$ and for $k \geq 1$, we have $p(n, k) = (n - k + 1) \cdot p(n, k - 1) = n \cdot p(n - 1, k - 1)$. (★)

Extended Newton's Binomial Theorem

Theorem 17: For all non-zero $n \in \mathbb{R}$, we have:

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k$$

Example: Let $n = 1/2$, then we have an identity for $\sqrt{1+x}$:

$$\sqrt{1+x} = \sum_{k=0}^{\infty} \binom{1/2}{k} x^k$$

To actually *use* this fact, we need some *lemma*...

Lemma 18: For any integer $n \geq 1$, we have:

$$\binom{1/2}{n} = (-1)^{n+1} \cdot \binom{2n-2}{n-1} \cdot \frac{1}{2^{2n-1}} \cdot \frac{1}{n}$$

Extended Newton's Binomial Theorem [2]

Proof: By induction on n .

Base: $n = 1$.

$$\binom{1/2}{1} = \frac{1/2}{1!} = \frac{1}{2} = 1 \cdot 1 \cdot \frac{1}{2} \cdot 1 = \underbrace{(-1)^2}_1 \cdot \underbrace{\binom{2-2}{1-1}}_1 \cdot \underbrace{\frac{1}{2^{2-1}}}_{\frac{1}{2}} \cdot \underbrace{\frac{1}{1}}_1$$

Induction step: n to $n + 1$ for $n > 1$. We use the recursion $(\star) p(n, k) = n \cdot p(n - 1, k - 1)$:

$$\begin{aligned} \binom{1/2}{n+1} &= \frac{p(1/2, n+1)}{(n+1)!} = \frac{(1/2 - (n+1) + 1) \cdot p(1/2, n)}{(n+1) \cdot n!} = -\frac{n - 1/2}{n+1} \binom{1/2}{n} \\ &\stackrel{\text{IH}}{=} -\frac{n - 1/2}{n+1} (-1)^{n+1} \cdot \binom{2n-2}{n-1} \cdot \frac{1}{2^{2n-1}} \cdot \frac{1}{n} \\ &= \frac{2n}{2n} \cdot \frac{2n-1}{2n} \cdot (-1)^{n+2} \cdot \binom{2n-2}{n-1} \cdot \frac{1}{2^{2n-1}} \cdot \frac{1}{n+1} \\ &= (-1)^{n+2} \cdot \underbrace{\frac{(2n-2)! \cdot (2n-1) \cdot (2n)}{(n-1)! \cdot (n-1) \cdot n \cdot n}}_{\binom{2n}{n}} \cdot \frac{1}{2^{2n+1}} \cdot \frac{1}{n+1} \end{aligned} \quad \square$$

Catalan Numbers

Proposition 19: Now we can expand $\sqrt{1+n}$ into the following series:

$$\sqrt{1+n} = \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \sum_{n=1}^{\infty} -2 \cdot \binom{2n-2}{n-1} \cdot (-1)^n \cdot \frac{1}{2^{2n}} \cdot \frac{1}{n} \cdot x^n$$

Example: Going back to the example with the number of well-formed paranthesis expressions, we get:

$$\begin{aligned} F(x) &= \frac{1 - \sqrt{1-4x}}{2x} = \frac{1}{2x} \sum_{n=1}^{\infty} 2 \cdot \binom{2n-2}{n-1} \cdot (-1)^n \cdot \frac{1}{2^{2n}} \cdot \frac{1}{n} \cdot (-4x)^n \\ &= \frac{1}{x} \sum_{n=1}^{\infty} \binom{2n-2}{n-1} \frac{1}{n} x^n = \sum_{n=0}^{\infty} \binom{2n}{n} \frac{1}{n+1} x^n \end{aligned}$$

The numbers $C_n := \binom{2n}{n} \frac{1}{n+1}$ are called *Catalan numbers*.

Catalan Numbers [2]

Catalan numbers $C_n = \frac{1}{n+1} \binom{2n}{n}$ count many combinatorial structures:

- well-formed parenthesis strings of length $2n$,
- full binary trees with $n + 1$ leaves,
- triangulations of a convex $(n + 2)$ -gon,
- monotone lattice paths from $(0, 0)$ to (n, n) not crossing the diagonal.

$$C_0 = 1, \quad C_1 = 1, \quad C_2 = 2, \quad C_3 = 5, \quad C_4 = 14, \quad C_5 = 42, \quad \dots$$

Recurrence Relations

“A certain man placed one pair of rabbits in a place enclosed on all sides by a wall. How many pairs of rabbits can be produced from that pair in one year?”

— Leonardo Fibonacci, Liber Abaci (1202)

Recurrence Relations

A *recurrence relation* defines a sequence by expressing each term via its predecessors. Solving one means finding a *closed-form* formula that avoids recursion.

Example: *Arithmetic progression* as a recurrence:

$$a_n = \begin{cases} a_0 = \text{const} & \text{if } n = 0 \\ a_{n-1} + d & \text{if } n > 0 \end{cases}$$

Solution by telescoping: $a_n = a_0 + n \cdot d$.

Recursive Structure

Many combinatorial objects have *recursive structure*: a large instance can be decomposed into smaller instances of the same problem. A recurrence captures this decomposition *exactly*.

Example: How many binary strings of length n contain *no two consecutive* 1s? Let $b(n)$ denote the count.

Consider the last bit of a valid string of length n :

- **Ends in 0**: the first $n - 1$ bits form any valid string of length $n - 1$, giving $b(n - 1)$ ways.
- **Ends in 1**: the previous bit must be 0, so the first $n - 2$ bits form any valid string of length $n - 2$, giving $b(n - 2)$ ways.

Therefore $b(n) = b(n - 1) + b(n - 2)$, with $b(1) = 2$ and $b(2) = 3$.

This is the *Fibonacci recurrence* with shifted initial conditions. Once we solve it (using the characteristic equation), we get a closed formula that avoids enumerating all 2^n strings.

Telescoping

Before reaching for characteristic equations, try the simplest approach: *unwind the recurrence*.

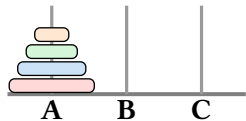
Example: Solve $T(n) = T(n - 1) + n$ with $T(0) = 0$.

Unwind by substitution:

$$\begin{aligned}T(n) &= T(n - 1) + n \\&= T(n - 2) + (n - 1) + n \\&= \vdots \\&= T(0) + 1 + 2 + \dots + n \\&= \frac{n(n + 1)}{2}\end{aligned}$$

Theorem 20: If $a_n = a_{n-1} + f(n)$, then $a_n = a_0 + \sum_{i=1}^n f(i)$ (telescoping sum).

Tower of Hanoi



Move n disks from peg A to peg C, one at a time, never placing a larger disk on a smaller one.

Strategy: move the top $n - 1$ disks to peg B, move the largest disk to peg C, then move the $n - 1$ disks from B to C.

Example: Let $T(n)$ be the minimum number of moves. Recurrence: $T(n) = 2T(n - 1) + 1$, with $T(1) = 1$.

Unwind:

$$\begin{aligned}T(n) &= 2T(n - 1) + 1 \\&= 2(2T(n - 2) + 1) + 1 = 4T(n - 2) + 3 \\&= 8T(n - 3) + 7 \\&= 2^k T(n - k) + (2^k - 1)\end{aligned}$$

Setting $k = n - 1$: $T(n) = 2^{n-1}T(1) + 2^{n-1} - 1 = 2^n - 1$.

Note: Telescoping works well for first-order recurrences. For higher orders and constant-coefficient recurrences, we need more systematic methods.

Linear Homogeneous Recurrence Relations

Definition 25: A *linear homogeneous* recurrence relation *of degree k* with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

where $c_1, c_2, \dots, c_k$ are constants (real or complex numbers), and $c_k \neq 0$.

Examples:

- $b_n = 2.71b_{n-1}$ is a linear homogeneous recurrence relation of degree 1.
- $F_n = F_{n-1} + F_{n-2}$ is a linear homogeneous recurrence relation of degree 2.
- The recurrence relation $a_n = a_{n-1} + a_{n-2}^2$ is *not linear*.
- The recurrence relation $H_n = 2H_{n-1} + 1$ is *not homogeneous*.

Characteristic Equations

Fix a linear homogeneous recurrence of degree k :

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} \quad (\star)$$

If we try a solution of the form $a_n = r^n$ (with $r \neq 0$), substituting into $(\star)$ and dividing by r^{n-k} yields:

Definition 26: The *characteristic equation* of $(\star)$ is:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$$

Its solutions $r_1, r_2, \dots$ are the *characteristic roots* of the recurrence. Each root yields a solution $a_n = r^n$.

Note: Why does this work? Substituting $a_n = r^n$ into the recurrence gives

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$$

which, after dividing by r^{n-k} (assuming $r \neq 0$), is exactly the characteristic equation. So r^n satisfies the recurrence *if and only if* r is a root. By linearity, any *linear combination* of such solutions is also a solution.

Distinct Roots Case

Theorem 21: Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1r - c_2 = 0$ has two *distinct* roots r_1 and r_2 . Then the sequence (a_n) is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if $a_n = \alpha_1r_1^n + \alpha_2r_2^n$ for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Proof: We prove both directions.

($\Rightarrow$) **Suppose** $a_n = \alpha_1r_1^n + \alpha_2r_2^n$. We verify it satisfies $a_n = c_1a_{n-1} + c_2a_{n-2}$. Since r_i is a root of $r^2 - c_1r - c_2 = 0$, we have $r_i^2 = c_1r_i + c_2$. Multiply both sides by r_i^{n-2} to get $r_i^n = c_1r_i^{n-1} + c_2r_i^{n-2}$. This holds for $i = 1$ and $i = 2$ separately, so taking a linear combination preserves equality:

$$\begin{aligned}c_1a_{n-1} + c_2a_{n-2} &= c_1(\alpha_1r_1^{n-1} + \alpha_2r_2^{n-1}) + c_2(\alpha_1r_1^{n-2} + \alpha_2r_2^{n-2}) \\&= \alpha_1(c_1r_1^{n-1} + c_2r_1^{n-2}) + \alpha_2(c_1r_2^{n-1} + c_2r_2^{n-2}) \\&= \alpha_1r_1^n + \alpha_2r_2^n = a_n.\end{aligned}$$

Distinct Roots Case [2]

($\Leftarrow$) **Suppose a_n satisfies the recurrence.** We show a_n *must* be of the form $\alpha_1 r_1^n + \alpha_2 r_2^n$. The recurrence and initial conditions a_0, a_1 determine the entire sequence: each a_n is computed from a_{n-1} and a_{n-2} . So it suffices to find α_1, α_2 such that $\alpha_1 r_1^n + \alpha_2 r_2^n$ matches the two initial values:

$$\begin{cases} \alpha_1 + \alpha_2 = a_0 \\ \alpha_1 r_1 + \alpha_2 r_2 = a_1 \end{cases}$$

This is a 2×2 linear system with determinant $r_2 - r_1$. Since $r_1 \neq r_2$, the determinant is nonzero, so a unique solution (α_1, α_2) exists. The sequence $\alpha_1 r_1^n + \alpha_2 r_2^n$ matches the initial conditions and satisfies the recurrence (by the first direction), so it *equals* a_n by uniqueness. $\square$

Worked Example

Example: Solve $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$.

- The *characteristic* equation is $r^2 - r - 2 = 0$.
- It has two *distinct* roots $r_1 = 2$ and $r_2 = -1$.
- The sequence (a_n) is a solution iff $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \dots$ and some constants α_1 and α_2 .

$$\begin{cases} a_0 = 2 = \alpha_1 + \alpha_2 \\ a_1 = 7 = \alpha_1 \cdot 2 + \alpha_2 \cdot (-1) \end{cases}$$

- Solving these two equations gives $\alpha_1 = 3$ and $\alpha_2 = -1$.
- Hence, the *solution* to the recurrence equation with given initial conditions is the sequence (a_n) with

$$a_n = 3 \cdot 2^n - (-1)^n$$

Fibonacci Numbers

Example: The recurrence $F_n = F_{n-1} + F_{n-2}$ has characteristic equation $r^2 - r - 1 = 0$ with roots:

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618, \quad \psi = \frac{1 - \sqrt{5}}{2} \approx -0.618.$$

The general solution is $F_n = \alpha_1 \varphi^n + \alpha_2 \psi^n$ for constants α_1, α_2 .

Using initial conditions $F_0 = 0$ and $F_1 = 1$:

$$\begin{cases} \alpha_1 + \alpha_2 = 0 \\ \alpha_1 \varphi + \alpha_2 \psi = 1 \end{cases}$$

Binet's Formula

Solving for α_1, α_2 yields $\alpha_1 = 1/\sqrt{5}$ and $\alpha_2 = -1/\sqrt{5}$, giving *Binet's formula*:

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$$

Since $|\psi| \approx 0.618 < 1$, the term $\psi^n \rightarrow 0$, so F_n is the nearest integer to $\varphi^n/\sqrt{5}$.

In particular, $F_{n+1}/F_n \rightarrow \varphi$ as $n \rightarrow \infty$.

n	1	2	3	4	5	6	7	8
F_n	1	1	2	3	5	8	13	21
F_{n+1}/F_n	1.000	2.000	1.500	1.667	1.600	1.625	1.615	1.619

Eigenvalues and the Companion Matrix

Definition 27: Let A be an $m \times m$ matrix. A scalar λ is an *eigenvalue* of A if there exists a nonzero vector $\vec{v}$ (called an *eigenvector*) such that

$$A\vec{v} = \lambda\vec{v}$$

The eigenvalues are the roots of the *characteristic polynomial* $p(\lambda) = \det(A - \lambda I)$.

Why does this matter for recurrences? Define the *state vector* $\vec{v}_n = (a_n, a_{n-1}, \dots, a_{n-k+1})^T$. Then the recurrence $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ is equivalent to $\vec{v}_n = C\vec{v}_{n-1}$, where C is the *companion matrix*:

$$C = \begin{pmatrix} c_1 & c_2 & c_3 & \dots & c_k \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}$$

Eigenvalues and the Companion Matrix [2]

Iterating gives $\vec{v}_n = C^n \vec{v}_0$, so computing a_n reduces to *matrix exponentiation*. The characteristic polynomial of C is exactly $p(\lambda) = \lambda^k - c_1 \lambda^{k-1} - \dots - c_k$, i.e. the same characteristic equation we solve for the recurrence.

For $k = 2$: $C = \begin{pmatrix} c_1 & c_2 \\ 1 & 0 \end{pmatrix}$, and $\det(C - \lambda I) = \lambda^2 - c_1 \lambda - c_2$.

Example: Fibonacci: $C = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, eigenvalues φ, ψ . Diagonalizing $C = PDP^{-1}$ yields $C^n = PD^nP^{-1}$, recovering Binet's formula. Matrix exponentiation computes F_n in $O(\log n)$ multiplications (binary exponentiation), vs. $O(n)$ for naive iteration.

Single (Repeated) Root Case

Theorem 22: Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1r - c_2 = 0$ has a *single (repeated)* root r_0 . A sequence (a_n) is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if $a_n = (\alpha_1 + \alpha_2n)r_0^n$ for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Why nr_0^n ? When the root is repeated, r_0^n alone only provides one linearly independent solution. The second solution nr_0^n arises by taking the *derivative* of r^n with respect to r and then evaluating at $r = r_0$:

$$\frac{d}{dr}(r^n) = nr^{n-1} \quad \longrightarrow \quad \frac{d}{dr}(r^n) \cdot r = nr^n$$

This is analogous to the resonance term $te^{\lambda t}$ in differential equations.

Single (Repeated) Root Case [2]

Example: Solve $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$ and $a_1 = 6$.

The characteristic equation is $r^2 - 6r + 9 = 0$ with a single (repeated) root $r_0 = 3$.

Hence, the solution is of the form $a_n = (\alpha_1 + \alpha_2 n) 3^n$.

$$\begin{cases} a_0 = 1 = \alpha_1 \\ a_1 = 6 = (\alpha_1 + \alpha_2) \cdot 3 \end{cases} \implies \begin{cases} \alpha_1 = 1 \\ \alpha_2 = 1 \end{cases}$$

Thus, the *solution* is $a_n = (1 + n) 3^n$.

General Case

Theorem 23: Let $c_1, \dots, c_k$ be real numbers. Suppose that the recurrence $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ has t distinct characteristic roots $r_1, \dots, r_t$ with multiplicities $m_1, \dots, m_t$, respectively, so that $m_i \geq 1$ for $i = 1, \dots, t$ and $m_1 + \dots + m_t = k$.

Then the general solution of the recurrence relation is given by

$$a_n = \sum_{i=1}^t \left(r_i^n \cdot \sum_{j=0}^{m_i-1} \alpha_{i,j} n^j \right)$$

where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

Example: Find generic solution for $a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3}$.

The characteristic equation $r^3 - 7r^2 + 16r - 12 = 0$ has $t = 2$ distinct roots: $r_1 = 2$ repeated $m_1 = 2$ times, and $r_2 = 3$ with multiplicity $m_2 = 1$. Hence, the solution is of the form

$$a_n = (\alpha_{1,0} + \alpha_{1,1}n) 2^n + \alpha_{2,0} 3^n$$

Complex Roots

When the characteristic equation has complex roots, they come in *conjugate pairs* $\rho e^{\pm i\theta}$ (for recurrences with real coefficients).

Definition 28: For conjugate roots $\rho e^{\pm i\theta}$, the real general solution is:

$$a_n = \rho^n (\alpha \cos(n\theta) + \beta \sin(n\theta))$$

This follows from Euler's formula: $\rho^n e^{in\theta} = \rho^n (\cos(n\theta) + i \sin(n\theta))$.

Complex roots produce *oscillating exponential* behavior: magnitude ρ^n grows (or decays), while $\cos(n\theta)$ and $\sin(n\theta)$ produce periodic oscillation with period $2\pi/\theta$.

Complex Roots Example

Example: Solve $a_n = 2a_{n-1} - 2a_{n-2}$ with $a_0 = 1$ and $a_1 = 1$.

Characteristic equation $r^2 - 2r + 2 = 0$, roots $r = 1 \pm i$.

Polar form: $1 + i = \sqrt{2}e^{i\pi/4}$, so $\rho = \sqrt{2}$ and $\theta = \pi/4$.

General solution: $a_n = (\sqrt{2})^n (\alpha \cos(n\pi/4) + \beta \sin(n\pi/4))$.

Initial conditions: $a_0 = 1 \rightarrow \alpha = 1$ and $a_1 = 1 \rightarrow \beta = 0$.

Therefore $a_n = (\sqrt{2})^n \cos(n\pi/4)$.

Root Types Summary

For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, the solution method is always the same four steps. Only the *form* depends on the roots.

Step	Distinct Real	Repeated	Complex
1. Char. eq.	$r^2 - c_1 r - c_2 = 0$	$r^2 - c_1 r - c_2 = 0$	$r^2 - c_1 r - c_2 = 0$
2. Roots	$r_1 \neq r_2$	r_0 (multiplicity 2)	$\rho e^{\pm i\theta}$
3. Form	$\alpha_1 r_1^n + \alpha_2 r_2^n$	$(\alpha_1 + \alpha_2 n) r_0^n$	$\rho^n (\alpha \cos n\theta + \beta \sin n\theta)$
4. Fit ICs	2 equations	2 equations	2 equations

Recurrences and Differential Equations

Linear recurrences are *discrete analogues* of linear differential equations. The entire theory mirrors the continuous case.

Aspect	Differential Equation	Recurrence
1st order	$y' = cy \rightarrow y_0 e^{ct}$	$a_n = ca_{n-1} \rightarrow a_0 c^n$
2nd order	$y'' + py' + qy = 0$	$a_n + pa_{n-1} + qa_{n-2} = 0$
Char. eq.	$\lambda^2 + p\lambda + q = 0$	$r^2 + pr + q = 0$
Double root	$(c_1 + c_2 t)e^{\lambda t}$	$(\alpha_1 + \alpha_2 n)r_0^n$
Particular sol.	Undetermined coefficients	Same method
Resonance	$te^{\lambda t}$ factors	nr_0^n factors

Solving Linear Recurrences

Step	Homogeneous	Non-homogeneous
1. Classify	No $F(n)$ term?	Has $F(n)$ term?
2. Char. eq.	$r^k - c_1 r^{k-1} - \dots - c_k = 0$	Same (ignore $F(n)$)
3. Roots	Distinct / repeated / complex	Same
4. Homog. sol.	$a_n^{(h)}$ from root type	Same
5. Partic. sol.	—	Guess form matching $F(n)$
6. General	$a_n = a_n^{(h)}$	$a_n = a_n^{(h)} + a_n^{(p)}$
7. Fit ICs	Solve linear system	Solve linear system

Particular solution cheat sheet: $F(n)$ = polynomial of degree $d \rightarrow$ try $c_0 + c_1 n + \dots + c_d n^d$.
 $F(n) = c \cdot r^n$, r not a root $\rightarrow$ try βr^n . $F(n) = c \cdot r^n$, r is a root (mult. m) $\rightarrow$ try $\beta n^m r^n$ (resonance).

Linear Non-Homogeneous Recurrence Relations

Definition 29: A *linear non-homogeneous* recurrence relation *of degree k* with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

where $c_1, c_2, \dots, c_k$ are constants and $F(n) \neq 0$ is a function of n (the *forcing term*).

The *associated homogeneous recurrence* is the relation with $F(n)$ removed:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

Example:

- $a_n = 3a_{n-1} + 2n$ – non-homogeneous, $F(n) = 2n$ (polynomial).
- $a_n = 2a_{n-1} + 3^n$ – non-homogeneous, $F(n) = 3^n$ (exponential).
- $H_n = 2H_{n-1} + 1$ – non-homogeneous, $F(n) = 1$ (constant).

Solving Non-Homogeneous Recurrence Relations

Theorem 24: If $(a_n^{(p)})$ is a *particular* solution of the non-homogeneous recurrence $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$, then *every solution* is of the form $a_n = a_n^{(p)} + a_n^{(h)}$, where $a_n^{(h)}$ is a solution of the associated homogeneous recurrence.

The recipe: (1) solve the homogeneous part, (2) find a particular solution by guessing a form matching $F(n)$, (3) add them, (4) fit initial conditions.

Example: Find all solutions of $a_n = 3a_{n-1} + 2n$ with $a_1 = 3$.

Homogeneous part: $a_n^{(h)} = \alpha 3^n$.

Particular solution: $F(n) = 2n$ is degree-1 polynomial, so try $p_n = cn + d$. Substituting: $cn + d = 3(c(n-1) + d) + 2n$. Equating coefficients: $c = -1$, $d = -\frac{3}{2}$. So $a_n^{(p)} = -n - \frac{3}{2}$.

General solution: $a_n = -n - \frac{3}{2} + \alpha 3^n$.

With $a_1 = 3$: $\alpha = \frac{11}{6}$, giving $a_n = -n - \frac{3}{2} + \left(\frac{11}{6}\right)3^n$.

Exponential Forcing Term

When $F(n) = c \cdot r^n$, try $a_n^{(p)} = \beta r^n$.

Warning: resonance. If r is already a characteristic root, the trial βr^n fails. Instead, multiply by n : try $a_n^{(p)} = \beta n r^n$ (or n^m if r has multiplicity m).

Example: Solve $a_n = 2a_{n-1} + 3^n$ with $a_0 = 1$.

- *Homogeneous solution:* $a_n^{(h)} = \alpha 2^n$.
- *Particular solution.* Since 3 is *not* a characteristic root, try $a_n^{(p)} = \beta \cdot 3^n$:

$$\beta 3^n = 2\beta 3^{n-1} + 3^n \implies \beta = 2\frac{\beta}{3} + 1 \implies \beta = 3$$

So $a_n^{(p)} = 3^{n+1}$.

- *General solution:* $a_n = \alpha 2^n + 3^{n+1}$.
- Using $a_0 = 1$: $1 = \alpha + 3$, so $\alpha = -2$.
- The *solution* is $a_n = 3^{n+1} - 2^{n+1}$.

Recurrences and Generating Functions

Both characteristic equations and generating functions solve the same recurrences, but differently.

Key idea. A generating function encodes the *entire sequence* (a_n) as coefficients of a power series $A(x) = \sum a_n x^n$. The recurrence turns into an *algebraic equation* for $A(x)$.

Example: *Derive* the generating function for Fibonacci: $F_n = F_{n-1} + F_{n-2}$, $F_0 = 0$, $F_1 = 1$.

Write $G(x) = F_0 + F_1 x + F_2 x^2 + F_3 x^3 + \dots = x + x^2 + 2x^3 + 3x^4 + \dots$

Multiply by x and x^2 to shift indices:

$$\begin{aligned} xG(x) &= \\ &= F_0 x + F_1 x^2 + F_2 x^3 + \dots + x^2 G(x) = \\ &= F_0 x^2 + F_1 x^3 + \dots \end{aligned}$$

Since $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$, each coefficient of $G - xG - x^2 G$ vanishes except $F_1 x = x$:

Recurrences and Generating Functions [2]

$$G(x) - xG(x) - x^2G(x) = x \implies G(x) = \frac{x}{1 - x - x^2}$$

Now factor the denominator. The roots of $1 - x - x^2 = 0$ are:

$$x = \frac{1 \pm \sqrt{5}}{-2} \implies x_1 = -\frac{1}{\varphi}, \quad x_2 = -\frac{1}{\psi}$$

where $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$.

The denominator $1 - x - x^2$ and the characteristic polynomial $r^2 - r - 1$ differ only by the substitution $x \rightarrow \frac{1}{r}$. Setting the denominator to zero gives values of x where the series *diverges* — these are the *poles* of $G(x)$. The characteristic roots r_1, r_2 are the *reciprocals* of these poles.

Recurrences and Generating Functions [3]

Partial fraction decomposition of $G(x)$ then produces exactly Binet's formula:

$$G(x) = \frac{x}{1-x-x^2} = \frac{1}{\sqrt{5}} \left(\frac{1}{1-\varphi x} - \frac{1}{1-\psi x} \right) = \frac{1}{\sqrt{5}} \sum_{n=0}^{\infty} (\varphi^n - \psi^n) x^n$$

Why two methods?

- **Characteristic equations** are mechanical and fast for constant-coefficient recurrences.
- **Generating functions** reveal combinatorial structure, handle variable coefficients, produce closed forms via partial fractions – at the cost of more algebra.

Summary: Recurrence Relations

Solution methods

1. *Telescoping* — unwind first-order recurrences
2. *Characteristic equations* — for constant-coefficient linear recurrences
3. *Generating functions* — encode the whole sequence algebraically

Recipe: characteristic equation

Polynomial $\rightarrow$ roots $\rightarrow$ general solution
(distinct / repeated / complex) $\rightarrow$ fit initial conditions.

General solution forms

- Distinct: $\alpha_1 r_1^n + \alpha_2 r_2^n$
- Repeated: $(\alpha_1 + \alpha_2 n) r_0^n$
- Complex: $\rho^n (\alpha \cos n\theta + \beta \sin n\theta)$
- Non-homogeneous: $a_n = a_n^h + a_n^p$

Connections

- Companion matrix, eigenvalues $\Rightarrow$ linear algebra
- Same structure as linear differential equations
- Characteristic roots = reciprocals of GF poles

Annihilators

Operators

Definition 30: *Operators* are higher-order functions that transform functions into other functions.

For example, differential and integral operators $\frac{d}{dx}$ and $\int dx$ are core operators in calculus.

In combinatorics, we are interested in the following three operators:

- *Sum*: $(f + g)(n) := f(n) + g(n)$
- *Scale*: $(\alpha \cdot f)(n) := \alpha \cdot f(n)$
- *Shift*: $(\mathbf{E} f)(n) := f(n + 1)$

Examples:

- Scale and Shift operators are *linear*: $\mathbf{E}(f - 3(g - h)) = \mathbf{E} f + (-3) \mathbf{E} g + 3 \mathbf{E} h$
- Operators are *composable*: $(\mathbf{E} - 2)f := \mathbf{E} f + (-2)f$
- $\mathbf{E}^2 f = \mathbf{E}(\mathbf{E} f)$
- $\mathbf{E}^k f(n) = f(n + k)$
- $(\mathbf{E} - 2)^2 = (\mathbf{E} - 2)(\mathbf{E} - 2)$
- $(\mathbf{E} - 1)(\mathbf{E} - 2) = \mathbf{E}^2 - 3 \mathbf{E} + 2$

Applying Operators

Examples: Below are the results of applying different operators to $f(n) = 2^n$:

$$2f(n) = 2 \cdot 2^n = 2^{n+1}$$

$$3f(n) = 3 \cdot 2^n$$

$$\mathbf{E} f(n) = 2^{n+1}$$

$$\mathbf{E}^2 f(n) = 2^{n+2}$$

$$(\mathbf{E} - 2)f(n) = \mathbf{E} f(n) - 2f(n) = 2^{n+1} - 2^{n+1} = 0$$

$$(\mathbf{E}^2 - 1)f(n) = \mathbf{E}^2 f(n) - f(n) = 2^{n+2} - 2^n = 3 \cdot 2^n$$

Compound Operators

The compound operators can be seen as polynomials in “variable” $\mathbf{E}$.

Example: The compound operators $\mathbf{E}^2 - 3\mathbf{E} + 2$ and $(\mathbf{E} - 1)(\mathbf{E} - 2)$ are equivalent:

$$\text{Let } g(n) := (\mathbf{E} - 2)f(n) = f(n + 1) - 2f(n)$$

$$\begin{aligned}\text{Then } (\mathbf{E} - 1)(\mathbf{E} - 2)f(n) &= (\mathbf{E} - 1)g(n) \\ &= g(n + 1) - g(n) \\ &= [f(n + 2) - 2f(n + 1)] - [f(n + 1) - 2f(n)] \\ &= f(n + 2) - 3f(n + 1) + 2f(n) \\ &= (\mathbf{E}^2 - 3\mathbf{E} + 2)f(n) \quad \checkmark\end{aligned}$$

Operators Summary

Operator	Definition
addition	$(f + g)(n) := f(n) + g(n)$
subtraction	$(f - g)(n) := f(n) - g(n)$
multiplication	$(\alpha \cdot f)(n) := \alpha \cdot f(n)$
shift	$\mathbf{E} f(n) := f(n + 1)$
k-fold shift	$\mathbf{E}^k f(n) := f(n + k)$
composition	$(\mathbf{X} + \mathbf{Y})f := \mathbf{X} f + \mathbf{Y} f$
	$(\mathbf{X} - \mathbf{Y})f := \mathbf{X} f - \mathbf{Y} f$
	$\mathbf{X} \mathbf{Y} f := \mathbf{X}(\mathbf{Y} f) = \mathbf{Y}(\mathbf{X} f)$
distribution	$\mathbf{X}(f + g) = \mathbf{X} f + \mathbf{X} g$

Annihilators

Definition 31: An *annihilator* of a function f is any non-trivial operator X such that $Xf = 0$.

Intuition: An annihilator “kills” the function — after applying the operator, nothing remains. This is the same idea as finding roots of a polynomial: if $(x - 2)$ is a factor, then plugging in $x = 2$ makes the polynomial zero. Here, $(\mathbf{E} - a)$ is a factor, and applying it to a^n makes the result zero.

Examples:

- $(\mathbf{E} - 1)$ annihilates any constant $f(n) = c$: $(\mathbf{E} - 1)c = c - c = 0$.
- $(\mathbf{E} - 2)$ annihilates $f(n) = \alpha \cdot 2^n$: $(\mathbf{E} - 2)(\alpha 2^n) = \alpha 2^{n+1} - 2\alpha 2^n = 0$.
- $(\mathbf{E} - 1)^2$ annihilates any linear $f(n) = \alpha n + \beta$: $(\mathbf{E} - 1)(\alpha n + \beta) = \alpha$ (a constant), and $(\mathbf{E} - 1)\alpha = 0$.

Annihilators Summary

Operator	Functions annihilated
$\mathbf{E} - 1$	α
$\mathbf{E} - a$	αa^n
$(\mathbf{E} - a)(\mathbf{E} - b)$	$\alpha a^n + \beta b^n$ [if $a \neq b$]
$(\mathbf{E} - a_0)(\mathbf{E} - a_1) \dots (\mathbf{E} - a_k)$	$\sum_{i=0}^k \alpha_i a_i^n$ [if a_i are distinct]
$(\mathbf{E} - 1)^2$	$\alpha n + \beta$
$(\mathbf{E} - a)^2$	$(\alpha n + \beta) a^n$
$(\mathbf{E} - a)^2(\mathbf{E} - b)$	$(\alpha n + \beta) a^n + \gamma b^n$ [if $a \neq b$]
$(\mathbf{E} - a)^d$	$(\sum_{i=0}^{d-1} \alpha_i n^i) a^n$

Properties of Annihilators

Theorem 25: If X annihilates f , then X also annihilates αf for any constant α .

Proof: Since operators are linear (they distribute over scaling), $X(\alpha f) = \alpha \cdot X f = \alpha \cdot 0 = 0$. □

Theorem 26: If X annihilates both f and g , then X also annihilates $f \pm g$.

Proof: By linearity of operators: $X(f \pm g) = X f \pm X g = 0 \pm 0 = 0$. □

Theorem 27: If X annihilates f , then X also annihilates $\mathbf{E} f$.

Proof: Any operator X that is a polynomial in $\mathbf{E}$ *commutes* with $\mathbf{E}$ itself.

Therefore $X(\mathbf{E} f) = \mathbf{E}(X f) = \mathbf{E}(0) = 0$. □

Properties of Annihilators [2]

Theorem 28: If X annihilates f and Y annihilates g , then XY annihilates $f \pm g$.

Proof: Since X annihilates f , and by the previous theorem X also annihilates $E f$, $E^2 f$, *etc.*, the operator X annihilates *any* function obtained by applying shift-operators to f . In particular, X annihilates $Y f$.

Now apply XY to $f \pm g$:

$$XY(f \pm g) = X(Y f \pm Y g) = X(Y f) \pm X(Y g)$$

The first term $X(Y f)$: since X annihilates f and commutes with Y , this is $Y(X f) = Y(0) = 0$. The second term $X(Y g)$: since Y annihilates g , we get $X(0) = 0$. Both terms are zero, so the result is zero. $\square$

Annihilating Recurrences

The annihilator method is a systematic five-step procedure for solving recurrences.

1. *Rewrite* the recurrence in operator form (replace a_n with $\mathbf{E}$ -operators).
2. *Identify* the annihilator of the *left-hand side* and annihilate the *residue* (right-hand side) by multiplying additional factors.
3. *Factor* the annihilator into linear factors $(\mathbf{E} - a)$.
4. *Read off* the generic solution from the factorization (one term per factor, using the summary table).
5. *Fit* the initial conditions to determine the free constants.

Example: $r(n) = 5r(n-1)$ with $r(0) = 3$.

Step 1. $r(n+1) - 5r(n) = 0$
 $(\mathbf{E} - 5)r(n) = 0$

Step 2. $(\mathbf{E} - 5)$ annihilates $r(n)$.

Thus, $r(n) = 3 \cdot 5^n$.

Step 3. $(\mathbf{E} - 5)$ is already factored.

Step 4. $r(n) = \alpha 5^n$ is a generic solution.

Step 5. $r(0) = \alpha = 3 \implies \alpha = 3$

Annihilating Recurrences [2]

Example: $T(n) = 2T(n-1) + 1$ with $T(0) = 0$

1. $T(n+1) - 2T(n) = 1$
 $(\mathbf{E}-2)T(n) = 1$
2. $(\mathbf{E}-2)$ does *not* annihilate $T(n)$: the residue is 1.
 $(\mathbf{E}-1)$ annihilates the residue 1.
Thus, $(\mathbf{E}-1)(\mathbf{E}-2)$ annihilates $T(n)$.
3. $(\mathbf{E}-1)(\mathbf{E}-2)$ is already factored.
4. $T(n) = \alpha 2^n + \beta$ is a generic solution.
5. Find the coefficients α, β using $T(0) = 0$ and $T(1) = 2T(0) + 1 = 1$:

$$\begin{cases} T(0) = 0 = \alpha \cdot 2^0 + \beta \\ T(1) = 1 = \alpha \cdot 2^1 + \beta \end{cases} \implies \begin{cases} \alpha = 1 \\ \beta = -1 \end{cases}$$

Thus, $T(n) = 2^n - 1$.

Annihilating Recurrences [3]

Example: $T(n) = T(n-1) + 2T(n-2) + 2^n - n^2$

1. Operator form:

$$(\mathbf{E}^2 - \mathbf{E} - 2)T(n) = \mathbf{E}^2(2^n - n^2)$$

2. Annihilator:

$$(\mathbf{E}^2 - \mathbf{E} - 2)(\mathbf{E} - 2)(\mathbf{E} - 1)^3$$

3. Factorization:

$$(\mathbf{E} + 1)(\mathbf{E} - 2)^2(\mathbf{E} - 1)^3$$

4. Generic solution:

$$T(n) = \alpha(-1)^n + (\beta n + \gamma)2^n + \delta n^2 + \varepsilon n + \zeta$$

5. There are no initial conditions. We can only provide an asymptotic bound.

Thus, $T(n) \in \Theta(n2^n)$

Annihilators and Characteristic Equations

Note: The characteristic equation method and the annihilator method are *the same algebra* in different notation. Replacing r with $\mathbf{E}$ in the characteristic polynomial gives the annihilator, and vice versa. Understanding both gives you flexibility: use whichever is more convenient.

Theorem 29: For the linear homogeneous recurrence $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$:

- **Characteristic polynomial:** $p(r) = r^k - c_1 r^{k-1} - \dots - c_k$
- **Annihilator polynomial:** $p(\mathbf{E}) = \mathbf{E}^k - c_1 \mathbf{E}^{k-1} - \dots - c_k$

If $r_1, \dots, r_t$ are the distinct characteristic roots with multiplicities $m_1, \dots, m_t$, then

$$p(r) = (r - r_1)^{m_1} \dots (r - r_t)^{m_t} = p(\mathbf{E})$$

The factorization is identical; only the interpretation differs.

Proof: The recurrence $a_n - c_1 a_{n-1} - \dots - c_k a_{n-k} = 0$ can be written as $(\mathbf{E}^k - c_1 \mathbf{E}^{k-1} - \dots - c_k) a_n = 0$. This is the *annihilator polynomial* applied to a_n .

Annihilators and Characteristic Equations [2]

The characteristic equation is $r^k - c_1 r^{k-1} - \dots - c_k = 0$.

Both polynomials have the same coefficients — the only difference is the symbol (r vs. $\mathbf{E}$). Therefore they have the same roots and the same factorization. □

Example: $a_n = 5a_{n-1} - 6a_{n-2}$.

- **Characteristic:** $r^2 - 5r + 6 = (r - 2)(r - 3) = 0$.
- **Annihilator:** $(\mathbf{E} - 2)(\mathbf{E} - 3)a_n = 0$.
- Both give: $a_n = \alpha 2^n + \beta 3^n$.

Common Mistakes

Example: Solve $a_n = 2a_{n-1} - 2^n$ with $a_0 = 1$.

Wrong answer: “Characteristic equation: $r - 2 = 0$, so $r = 2$. General solution: $a_n = c \cdot 2^n$.”

Where’s the error?

- This is *non-homogeneous* (has extra term “ -2^n ”), but the student *ignored it!*
- We need a *particular* solution in addition to the homogeneous one.
- Since 2 is *also* the characteristic root (resonance!), the simple trial $\beta 2^n$ fails. We must try $a_n^{(p)} = \beta n \cdot 2^n$.

Correct: $a_n = (\alpha + n)2^n$. Using $a_0 = 1$: $\alpha = 1$, so $a_n = (1 + n)2^n$.

Summary: Annihilators

Core idea. An operator X *annihilates* f if $X f = 0$.

The factor $(\mathbf{E} - a)$ kills a^n . Higher powers $(\mathbf{E} - a)^d$ kill $(\alpha_0 + \alpha_1 n + \dots + \alpha_{d-1} n^{d-1}) a^n$.

Finding the annihilator of a recurrence *automatically* reveals the form of the solution.

Algorithm

1. Rewrite recurrence in operator form
2. Find the annihilator of both sides
3. Factor into $(\mathbf{E} - a_i)$ terms
4. Read off generic solution from factors
5. Fit initial conditions

Key properties

- If X annihilates f ,
then it also annihilates $\alpha f, f \pm g, \mathbf{E} f$
- If X annihilates f and Y annihilates g ,
then $X Y$ annihilates $f \pm g$
- Same polynomial as characteristic equation, with
 $r \Rightarrow \mathbf{E}$

Asymptotic Analysis

*“A mathematician who is not also something of a poet
will never be a complete mathematician.”*

— Karl Weierstrass

Asymptotic Notation

Definition 32 (*Big-O notation*): $f(n) \in O(g(n))$ means f is *asymptotically bounded from above* by g , up to a constant factor:

$$f(n) \in O(g(n)) \iff \exists c > 0. \exists n_0. \forall n > n_0 : |f(n)| \leq c \cdot g(n)$$

Definition 33 (*Small-o notation*): $f(n) \in o(g(n))$ means f is *strictly dominated* by g : no constant makes f keep up with g .

$$f(n) \in o(g(n)) \iff \forall c > 0. \exists n_0. \forall n > n_0 : |f(n)| \leq c \cdot g(n)$$

Note: The only difference is $\exists c$ (Big-O: *some* constant works) vs. $\forall c$ (Small-o: *every* constant works, no matter how small).

Note: Flip $\leq$ to $\geq$ to get the dual notations $f \in \Omega(g)$ (bounded below) and $f \in \omega(g)$ (strictly dominates).

Asymptotic Notation [2]

Definition 34 (*Theta notation*): $f \in \Theta(g)$ iff $f \in O(g)$ and $f \in \Omega(g)$ – f and g grow at the *same rate*, up to constant factors.

Example: Prove that $3n^2 + 5n \in \Theta(n^2)$.

Upper bound: $3n^2 + 5n \leq 3n^2 + 5n^2 = 8n^2$ for $n \geq 1$, so $c = 8$, $n_0 = 1$.

Lower bound: $3n^2 + 5n \geq 3n^2$ for $n \geq 0$, so $c = 3$, $n_0 = 0$.

Both bounds hold, so $3n^2 + 5n \in \Theta(n^2)$.

Limits

Notation	Name	Description	Limit definition
$f \in o(g)$	Small Oh	f is dominated by g	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$
$f \in O(g)$	Big Oh	f is bounded above by g	$\limsup_{n \rightarrow \infty} \frac{ f(n) }{g(n)} < \infty$
$f \sim g$	Equivalence	f is asymptotically equal to g	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$
$f \in \Omega(g)$	Big Omega	f is bounded below by g	$\liminf_{n \rightarrow \infty} \frac{f(n)}{g(n)} > 0$
$f \in \omega(g)$	Small Omega	f dominates g	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$

Asymptotic Equivalence

Definition 35: $f \sim g$ means f and g are *asymptotically equivalent* — their ratio tends to 1:

$$f \sim g \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$$

Example: $n^2 + 3n \sim n^2$, since $\frac{n^2+3n}{n^2} = 1 + \frac{3}{n} \rightarrow 1$.

Note: $f \sim g$ is *stronger* than $f \in \Theta(g)$: $2n \in \Theta(n)$ but $2n \sim n$ is false (the ratio is 2, not 1).

Properties of Asymptotics

Theorem 30: The asymptotic notations satisfy:

$$f \in O(g) \text{ and } f \in \Omega(g) \iff f \in \Theta(g)$$

$$f \in O(g) \iff g \in \Omega(f)$$

$$f \in o(g) \iff g \in \omega(f)$$

$$f \in o(g) \rightarrow f \in O(g)$$

$$f \in \omega(g) \rightarrow f \in \Omega(g)$$

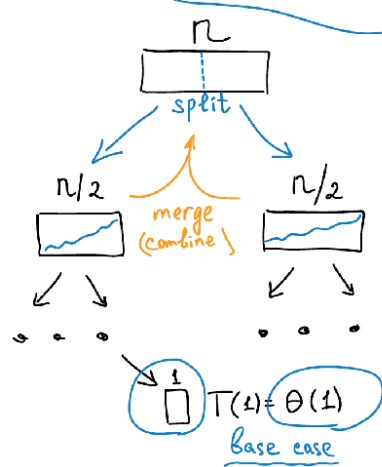
$$f \sim g \rightarrow f \in \Theta(g)$$

Note: Growth rate hierarchy: $1 \subset o(\ln n) \subset o(n) \subset o(n \ln n) \subset o(n^2) \subset o(2^n) \subset o(n!)$.

Each class is strictly contained in the next: $n^k \in o(c^n)$ for any k and $c > 1$.

Divide-and-Conquer Algorithms Analysis

Divide-and-Conquer



$$T(n) = \underbrace{2 T(n/2)}_{\text{recursive work}} + \underbrace{\Theta(n)}_{\text{split/merge work}}$$

$$\left\{ \begin{aligned} \Sigma &= 2 \cdot 2 \cdot \underbrace{T(n/4)}_{\text{merge}} = n \end{aligned} \right.$$

$$\left\{ \begin{aligned} \Sigma &\approx 2^{i+1} T(n/2^{i+1}) = n \end{aligned} \right.$$

$$\left\{ \begin{aligned} \Sigma &= n \cdot \Theta(1) = n \end{aligned} \right.$$

$$\underbrace{\Theta(n)}_{\text{work on each level of recursion tree}} \cdot \underbrace{\Theta(\log n)}_{\text{height of recursion tree}} = \Theta(n \log n)$$

Divide-and-Conquer Recurrence

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

- $T(n)$ is the *cost* of the recursive algorithm
- a is the number of *parts* (*sub-problems*)
- n/b is the *size* of each part
- $T\left(\frac{n}{b}\right)$ is the cost of each *sub-problem*
- $f(n)$ is the cost of *splitting* and *merging* the solutions of the subproblems

Hereinafter, $c_{\text{crit}} = \log_b a$ is a *critical constant*.

Master Theorem

The master theorem [2] gives a direct asymptotic bound for recurrences of the form

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

where $c_{\text{crit}} = \log_b a$ is the *critical exponent*.

Intuition: The recurrence splits into a sub-problems of size n/b . The total work at each level of the recursion tree is a copies of the sub-problem cost, plus the merge cost $f(n)$.

- If $f(n)$ is small relative to $n^{c_{\text{crit}}}$, the recursion *leaves* dominate — total cost is $\Theta(n^{c_{\text{crit}}})$.
- If $f(n)$ is comparable to $n^{c_{\text{crit}}}$, the levels contribute equally — a log factor appears.
- If $f(n)$ is large, the *root level* dominates — total cost is $\Theta(f(n))$.

Master Theorem [2]

Case	Description	Condition	Bound
Case I	“merge” $\ll$ “recursion”	$f(n) \in O(n^c), c < c_{\text{crit}}$	$T(n) \in \Theta(n^{c_{\text{crit}}})$
Case II	“merge” $\approx$ “recursion”	$f(n) \in \Theta(n^{c_{\text{crit}}} \log^k n), k \geq 0$	$T(n) \in \Theta(n^{c_{\text{crit}}} \log^{k+1} n)$
Case III	“merge” $\gg$ “recursion”	$f(n) \in \Omega(n^c), c > c_{\text{crit}}$	$T(n) \in \Theta(f(n))$

Note: Case III also requires the *regularity condition* to hold: $af(n/b) \leq kf(n)$ for some constant $k < 1$ and all sufficiently large n .

Examples of Master Theorem Application

Example: $T(n) = 2T(n/4) + n^{0.51}$

$a = 2, b = 4, c_{\text{crit}} = \log_4 2 = \frac{1}{2}$. $f(n) = n^{0.51} \in \Omega(n^c)$ with $c = 0.51 > 0.5 = c_{\text{crit}}$.

Case III: $T(n) \in \Theta(n^{0.51})$.

Example: $T(n) = 5T(n/25) + n^{0.49}$

$a = 5, b = 25, c_{\text{crit}} = \log_{25} 5 = \frac{1}{2}$. $f(n) = n^{0.49} \in O(n^c)$ with $c = 0.49 < 0.5 = c_{\text{crit}}$.

Case I: $T(n) \in \Theta(n^{\frac{1}{2}})$.

Example: $T(n) = 3T(n/9) + \sqrt{n}$

$a = 3, b = 9, c_{\text{crit}} = \log_9 3 = \frac{1}{2}$. $f(n) = n^{\frac{1}{2}} = n^{c_{\text{crit}}}$, so $f(n) \in \Theta(n^{c_{\text{crit}}} \log^0 n)$.

Case II ($k = 0$): $T(n) \in \Theta(n^{\frac{1}{2}} \log n)$.

Akra–Bazzi Method

The Akra–Bazzi method [3] is a *generalization* of the master theorem to recurrences of the form

$$T(n) = f(n) + \sum_{i=1}^k a_i T\left(b_i n + \underbrace{h_i(n)}_{*}\right)$$

- k is a constant
- $a_i > 0$
- $0 < b_i < 1$
- $h_i(n) \in O\left(\frac{n}{\log^2 n}\right)$ is a *small perturbation*

Bound of $T(n)$ by Akra–Bazzi method:

$$T(n) \in \Theta\left(n^p \cdot \left(1 + \int_1^n \frac{f(x)}{x^{p+1}} dx\right)\right)$$

where p is the solution for the equation $\sum_{i=1}^k a_i b_i^p = 1$

Akra–Bazzi Example

Example: Suppose the runtime of an algorithm is expressed by the following recurrence relation:

$$T(n) = \begin{cases} 1 & \text{for } 0 \leq n \leq 3 \\ n^2 + \frac{7}{4}T(\lfloor \frac{1}{2}n \rfloor) + T(\lceil \frac{3}{4}n \rceil) & \text{for } n > 3 \end{cases}$$

- Note that the Master Theorem *is not* applicable here, since there are *two* different recursive terms.
- Let's apply the Akra–Bazzi method. First, solve the equation $\frac{7}{4}(\frac{1}{2})^p + (\frac{3}{4})^p = 1$. This gives us $p = 2$.
- Next, use the formula from AB-method to obtain the bound:

$$\begin{aligned} T(x) &\in \Theta\left(x^p \left(1 + \int_1^x \frac{f(u)}{u^{p+1}} du\right)\right) = \\ &= \Theta\left(x^2 \left(1 + \int_1^x \frac{u^2}{u^3} du\right)\right) = \\ &= \Theta(x^2(1 + \ln x)) = \\ &= \Theta(x^2 \log x) \end{aligned}$$

Summary: Asymptotic Analysis

Notations

- $O(g)$: bounded above by g (some constant c)
- $\Omega(g)$: bounded below by g
- $\Theta(g)$: bounded above *and* below — tight
- $o(g)$: strictly dominated (no constant saves you)
- $\omega(g)$: strictly dominates
- $f \sim g$: ratio $\rightarrow 1$ (stronger than Θ)

Growth hierarchy: $1 \subset o(\ln n) \subset o(n) \subset o(n \ln n) \subset o(n^2) \subset o(2^n) \subset o(n!)$

Master Theorem

$T(n) = aT(n/b) + f(n)$, where $c_{\text{crit}} = \log_b a$.

Compare $f(n)$ with $n^{c_{\text{crit}}}$:

- Case I: f is smaller $\rightarrow \Theta(n^{c_{\text{crit}}})$
- Case II: f is comparable $\rightarrow \Theta(n^{c_{\text{crit}}} \log n)$
- Case III: f dominates $\rightarrow \Theta(f(n))$

Akra-Bazzi

Generalizes Master Theorem to unequal subproblem sizes. Solve $\sum a_i b_i^p = 1$ for p , then integrate.

Advanced Topics

Gamma Function

Definition 36: *Gamma function* $\Gamma(x)$ is the most common *extension* of the factorial function to real and complex numbers. It is defined for all complex numbers $z \in \mathbb{C}$ (except non-positive integers) as

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

For positive integers $z = n$, it is defined as

$$\Gamma(n) = (n-1)!$$

Motivation: The factorial is defined for positive integers as $n! = 1 \cdot 2 \cdot \dots \cdot n = (n-1)! \cdot n$. We want to *extend* this definition to *all real numbers* and capture its *recursive* nature. Specifically, we seek a smooth function $\Gamma(x)$ such that:

- $\Gamma(n+1) = n!$ for all $n \in \mathbb{N}$, matching the factorial.
- $\Gamma(x+1) = x \cdot \Gamma(x)$, satisfying a recursive property.
- $\Gamma(x)$ is defined for all real numbers $x > 0$.

Alternative Definitions

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

Gauss proposed a function $\Gamma(x)$ defined by the *limit*

$$\Gamma(x) := \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{x \cdot (x+1) \cdot \dots \cdot (x+n)} = \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{\prod_{k=0}^n (x+k)} \quad \text{for } x > 0$$

Integral Definition

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

Let's check that the integral definition is indeed a suitable definition of a gamma function.

$$\begin{aligned}\Gamma(z+1) &= \int_0^{\infty} t^z e^{-t} dt \\ &= [-t^z e^{-t}]_0^{\infty} + \int_0^{\infty} z t^{z-1} e^{-t} dt \\ &= \lim_{t \rightarrow \infty} (-t^z e^{-t}) - (-0^z e^{-0}) + z \int_0^{\infty} t^{z-1} e^{-t} dt\end{aligned}$$

Note that $-t^z e^{-t} \rightarrow 0$ as $t \rightarrow \infty$, so:

$$\Gamma(z+1) = z \int_0^{\infty} t^{z-1} e^{-t} dt = z \cdot \Gamma(z)$$

Limit Definition

$$\Gamma(x) = \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{\prod_{k=0}^n (x+k)}$$

Let's check that the limit definition is indeed a suitable definition of a gamma function.

Step 1. Write $\Gamma(x+1)$.

$$\Gamma(x+1) = \lim_{n \rightarrow \infty} \frac{n! \cdot n^{x+1}}{\prod_{k=0}^n (x+1+k)} = \lim_{n \rightarrow \infty} \frac{n! \cdot n^{x+1}}{\prod_{k=1}^{n+1} (x+k)}$$

Step 2. Multiply both numerator and denominator by x and rearrange:

$$= \lim_{n \rightarrow \infty} \frac{n! \cdot n^{x+1}}{(x+1) \cdot \dots \cdot (x+n+1)} = \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{x \cdot (x+1) \cdot \dots \cdot (x+n)} \cdot \frac{n}{x+n+1} \cdot x$$

Step 3. Take the limit. As $n \rightarrow \infty$, the ratio $\frac{n}{x+n+1}$ approaches 1.

$$\Gamma(x+1) = x \cdot \Gamma(x)$$

Equivalence of Definitions

Let's prove the equivalence of two definitions: integral and limit.

We claim:

$$\int_0^n t^{x-1} \left(1 - \frac{t}{n}\right)^n dt \stackrel{?}{=} \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{x \cdot (x+1) \cdot \dots \cdot (x+n)}$$

Note that as $n \rightarrow \infty$, the integrand $\left(1 - \frac{t}{n}\right)^n$ approaches e^{-t} , so this integral approximates $\Gamma(x)$.

Substitute $u = \frac{t}{n}$:

$$\int_0^n t^{x-1} \left(1 - \frac{t}{n}\right)^n dt = n^x \int_0^1 u^{x-1} (1-u)^n du = n^x \cdot B(x, n+1)$$

where $B(x, n+1)$ is the Beta function:

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x) \cdot \Gamma(y)}{\Gamma(x+y)}$$

Equivalence of Definitions [2]

Then:

$$I_n = n^x \cdot B(x, n+1) = n^x \cdot \frac{\Gamma(x) \cdot \Gamma(n+1)}{\Gamma(x+n+1)} = \frac{n! \cdot n^x}{x \cdot (x+1) \cdot \dots \cdot (x+n)}$$

Take the limit on both sides. Since $\lim_{n \rightarrow \infty} I_n = \Gamma(x)$, we have:

$$\Gamma(x) = \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{x \cdot (x+1) \cdot \dots \cdot (x+n)}$$

Gamma Function Applications

The Gamma function provides closed forms for several important combinatorial expressions.

- **Factorial extension:**

$$n! = \Gamma(n + 1), \quad \text{e.g. } \Gamma(1/2) = \sqrt{\pi}$$

Extends factorial to non-integer arguments.

- **Generalized binomial coefficients** (for arbitrary $r \in \mathbb{R}$):

$$\binom{r}{k} = \frac{\Gamma(r + 1)}{\Gamma(k + 1) \cdot \Gamma(r - k + 1)}$$

This is the definition underlying Newton's Binomial Theorem for real exponents.

- **Stirling's approximation:**

$$n! = \Gamma(n + 1) \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad \text{as } n \rightarrow \infty$$

Gamma Function Applications [2]

Useful for estimating $\binom{2n}{n}$, entropy bounds, and worst-case algorithm analysis.

- **Beta function:**

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \int_0^1 t^{x-1}(1-t)^{y-1}dt$$

Γ shows up whenever factorials, binomial coefficients, or power series hit non-integer parameters: probability distributions (Beta, Gamma, Student's t), analytic combinatorics, and elsewhere.

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